Higher Education or Just Higher? The Unintended Effects of Recreational Marijuana Legalization

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Abstract

This paper examines the effects of recreational marijuana legalization (RML) on educational attainment in the United States. Exploiting variation from the staggered adoption of RML across states, I find that RML reduces the probability of earning a bachelor's degree within four, five, and six years by 3.8%, 2.0%, and 3.1%, respectively, relative to pre-treatment means. In addition, RML lowers the three-year graduation rate at two-year colleges by 6.0% relative to the pre-treatment mean. These effects suggest that RML not only delays degree completion but may also lower educational attainment in the longer run. I further find that RML reduces grade point average (GPA) among currently enrolled college students ages 18–24 and study time among individuals in the same age range. Increases in marijuana-use frequency and other substance use among college-age individuals are plausible mechanisms, although the data do not establish their contribution to the educational effects. These findings highlight the potential educational costs of marijuana legalization alongside the documented benefits identified in prior research.

1 Introduction

Policies regulating potentially harmful and addictive substances involve a trade-off between consumer freedom and the negative spillovers such consumption may impose on others and on consumers' future selves. Marijuana remains illegal under federal law and is the most commonly used federally prohibited drug in the United States. At the state level, however, marijuana policy has liberalized substantially in recent decades, and because enforcement and access are largely determined by state legal status, these state-level changes are especially relevant

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for individual behavior.[1] Between 2012 and 2023, 25 U.S. states adopted laws legalizing recreational marijuana. These changes may have important consequences for decisions and behaviors related to higher education.

Higher educational attainment is strongly associated with labor market success, including higher wages, greater employment stability, and upward mobility, as well as non-economic benefits such as better health and civic participation. The relevance of marijuana policy for higher education is especially pronounced because marijuana use is highly prevalent during the college-going years. Nationally representative evidence from the National Survey on Drug Use and Health shows that 36.6% of young adults aged 18–25 used marijuana in the past year during 2022–2024, a rate substantially higher than that of older adults. This high prevalence during the college-going years coincides with a period in which individuals make consequential human-capital investment decisions. Although a large literature shows that substance use, including marijuana use, is associated with poorer academic performance (Fergusson, Horwood and Beautrais (2003); Cobb-Clark et al. (2015); Silins et al. (2015); Beverly, Castro and Opara (2019); Nawi et al. (2021); Nock, Minnes and Alberts (2017); Welsh, Shentu and Sarvey (2019)), yet credible causal evidence on the effects of RML on college attainment remains limited. By increasing access to marijuana and potentially altering consumption patterns among young adults, RML may have important implications for human-capital accumulation.

In this paper, I examine how RML affects educational attainment in the United States using a difference-in-differences framework. I estimate effects on institutional graduation outcomes, including bachelor's degree completion rates and graduation rates at two-year institutions. I consider three interrelated channels through which RML could reduce college completion: lower GPA, less time spent studying, and increased use of other drugs. I draw on three data sources: institution-level graduation-rate data from the Integrated Postsecondary Education Data System (IPEDS) to measure graduation outcomes, the American Time Use Survey (ATUS) to study students' time allocation, and the Panel Study of Income Dynamics (PSID) to examine GPA, current college enrollment, and substance use. The analysis exploits cross-state variation in the timing of legalization. I implement the estimator proposed by Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group.[2] A key identifying assumption is that, absent legalization, treated and comparison units would

[1] Chambers et al. (2023) document a steady decline in perceived risks associated with marijuana use. For example, the share of adults who consider daily cannabis smoking safer than tobacco increased from 36.7% in 2017 to 44.3% in 2021.

[2] In the Callaway and Sant'Anna (2021) framework, the not-yet-treated comparison group also includes units that are never treated.

have followed similar outcome trends. Consistent with this assumption, I find no systematic relationship between legalization timing and pre-treatment outcome measures.[3] In addition, contemporaneous accounts suggest that marijuana legalization efforts were widely viewed as difficult to predict (Bednarek, Doremus and Stith 2022).

Estimating the effect of RML on educational attainment is challenging for at least two reasons. First, RML may change the composition of students attending a given college through its effects on applications and enrollment. Blake, Thomas and Hess (2024) show that RML increases college applications, although the distribution of applicants' composite SAT scores changes little. To mitigate possible effects of legalization on selection into college (and college location), I focus on students who were already enrolled in college before their state legalized recreational marijuana. This rules out changes in initial college entry as a channel. Second, treated states legalized medical marijuana before adopting RML. I restrict the analysis to states that had implemented medical marijuana legalization (MML) by a common cutoff date, so that subsequent policy variation reflects RML adoption rather than differences in prior MML status.[4]

I find statistically significant and economically important adverse effects on higher-education outcomes. Among first-time, full-time students already enrolled at legalization, RML reduces BA completion at four-year institutions within four, five, and six years by 1.6, 1.1, and 1.8 percentage points, respectively. These declines equal 3.8%, 2.0%, and 3.1% of the corresponding pre-treatment means. The effects persist through the six-year horizon. RML also reduces graduation from two-year institutions within three years by 2.0 percentage points, or 6.0% of the pre-treatment mean.[5]

RML also affects intermediate academic outcomes, including students' GPA and the allocation of time between studying and leisure. Among currently enrolled college students ages 18–24, I find that RML reduces GPA by 10.5%. Using the shorter 2011–2015 enrollment series, I find no statistically significant effect on current college enrollment in the same age range. Extending the analysis of Chu and Gershenson (2018) to RML, I examine time allocation using a more detailed breakdown of study and leisure activities. RML reduces the time college-age individuals spend studying and increases the time they allocate to

[3]See Supplemental Appendix A2 and Table A1.

[4]Prior work finds that MML reduces education-related time and educational attainment (Chu 2015; Plunk et al. 2016). These effects complicate the interpretation of RML estimates when policy regimes overlap; many difference-in-differences designs are developed for a single treatment (Roth et al. 2023).

[5]Related evidence from Marie and Zölitz (2017) and Wright and Krieg (2020) documents adverse academic effects among students at Maastricht University and Western Washington University, respectively.

leisure. These effects are larger in magnitude than those associated with MML. This pattern is consistent with the idea that RML has a broader effect on marijuana use in the population than MML, which may in turn generate stronger behavioral responses. The substance-use results show increases in marijuana-use frequency and in cocaine use and frequency scores for amphetamines, barbiturates, and tranquilizers. I also find increases in the probability of ever drinking alcohol and smoking cigarettes.

Many studies have examined the relationship between marijuana consumption and the use of other dangerous substances. Here, I estimate the effects of RML on other substance use with the same difference-in-differences framework applied to educational outcomes. The available data do not establish that increased use of other drugs causes the estimated decline in educational attainment. Nevertheless, the substance-use estimates support it as a plausible mechanism.

This paper contributes to the literature in several ways. First, it quantifies the impact of RML on college completion, using the best data and methods available today. Second, it provides new evidence on mechanisms. Third, the paper distinguishes the effects of RML and MML, and compares them. These contributions complement a vast existing literature on consequences of marijuana legalization finding. For example, prior studies find that marijuana legalization reduces fertility (Papich 2022) and drug-related arrests (Sabia et al. 2021), increases opioid-related mortality (Mathur and Ruhm 2023), and raises marijuana consumption (Cerdá et al. 2020; Baggio, Chong and Simon 2020; Papich 2022; Muratori et al. 2023; Mauro et al. 2019). The evidence on employment and wages suggests more limited effects (Muratori et al. 2023).

The remainder of the paper is organized as follows. Section 2 provides background on marijuana laws. Section 3 describes the data. Section 4 presents the effects of RML on graduation rates. Section 5 examines GPA and time use, and Section 6 reports substance use and other health behavior. Section 7 presents robustness checks. Section 8 discusses the findings, and Section 9 concludes.

2 Marijuana Legalization

Figure 1 illustrates changes over time in the number of states that implemented medical marijuana legalization, opened medical marijuana dispensaries, legalized recreational marijuana, and opened recreational marijuana dispensaries. Table A2 summarizes the states and dates of marijuana legalization for medical and recreational purposes. These policies were adopted sequentially: recreational

marijuana legalization was implemented only after medical marijuana legalization.

The Controlled Substances Act of 1970 classified marijuana as a Schedule I drug, placing it among substances deemed to have a high potential for abuse and no accepted medical use under federal law (Svrakic et al. 2012; Mead 2019; Martins et al. 2022). Beginning in the 1970s, some states reduced criminal penalties for possession of small amounts of marijuana without creating legal commercial markets (Pacula and Smart 2017; Carliner et al. 2017).

In 1996, California became the first U.S. state to legalize medical marijuana. MML allows patients and their primary caregivers to possess and cultivate marijuana for medical purposes. In 2012, Washington and Colorado became the first states to legalize recreational marijuana. RML allows individuals aged 21 and older to possess, cultivate, and consume marijuana for recreational use. Under both MML and RML, states may authorize dispensaries. Before the recent wave of legalization, even possession of small amounts of marijuana could result in misdemeanor charges, fines, mandatory drug education, and, in many states, incarceration, while more serious offenses were often treated as felonies. These reforms therefore marked a departure from earlier criminal penalties for marijuana possession and distribution.

3 Data

Table 1 presents summary statistics for the main datasets. Table A3 reports summary statistics for additional health-related behaviors, including drinking, smoking, and other substance use. Both tables report overall means and standard deviations, as well as pre-treatment and post-treatment means.

This study draws on several data sources. The primary institution-level graduation outcomes are constructed from the IPEDS and cover entering cohorts from 1996 to 2017. It also reports detailed graduation outcomes, including four-year, five-year, and six-year bachelor's degree completion rates for four-year institutions and three-year graduation rates for two-year institutions.

To examine intermediate effects on students' time allocation, I use the ATUS data, an annual, nationally representative repeated cross-sectional survey.

In addition, I use the Transition into Adulthood Supplement (TAS) of the PSID to examine the effects of RML on GPA, college enrollment, substance use and other health behavior.[6]

[6]The substance-use analysis covers marijuana, cocaine, Group A (amphetamines, barbiturates, and tranquilizers), and Group B (diet pills and steroids). Some questions include prescribed use, so these measures do not identify nonmedical or illegal use.

3.1 Constructing the Sample

I restrict the analysis to states whose MML treatment year falls before a given cutoff year.[7] For each specification, I select a cutoff year and retain only those states that had already adopted MML by that year. The main results of the paper, including those on graduation rates, GPA, time use, substance use, and other health behaviors, are based on the sample defined by a 2010 cutoff year.[8] The primary motivation for this restriction is to better isolate the effects of RML from those of MML. It also limits the comparison to states that either eventually adopted RML or were plausibly closer to doing so, thereby improving the comparability of treated and control states and strengthening the plausibility of the parallel trends assumption. Table 2 lists the states included in the main analysis. The baseline sample contains 15 states, of which all but Hawaii had adopted RML by 2023.

For each Title IV four-year institution, I construct four-year, five-year, and six-year graduation rates as the shares of first-time, full-time students in an entering cohort who earn a bachelor's degree within four, five, and six years of initial enrollment, respectively. For Title IV two-year institutions, I construct the three-year graduation rate as the share of first-time, full-time students in an entering cohort who complete their program within three years. These measures adjust for transfer-outs, transfer-ins, and IPEDS-defined exclusions.[9]

For the main analysis, I examine cohorts entering between 2006 and 2017 for bachelor's degree outcomes. Using the 2006 cohort as the starting point ensures that the sample begins four years before the 2010 cutoff year. As a result, by the time four-year, five-year, and six-year completion outcomes are measured, all cohorts in the sample are already in states that had adopted MML. This helps separate the effects of MML from those of RML. For two-year institutions, I instead examine cohorts entering between 2008 and 2017.[10] To reduce sensitivity to very small cohort sizes—where graduation rates of 0 or 100 percent may arise simply because an entering class contains only a few students—the primary analysis uses a balanced panel of institutions and excludes an institution if its completion rate is either 0 or 100 percent for any cohort observed in the sample.[11]

[7] A detailed definition of the treatment year is provided in Section 4.1.

[8] Corresponding results based on alternative cutoff years are reported in Tables A4–A10.

[9] Appendix A1 lists these exclusions.

[10] The 2008 entering cohort is the earliest cohort for which complete graduation-rate information is available in IPEDS for two-year institutions.

[11] These cases typically arise when cohort size is very small or when the institution is inactive. I also verify through regression-based tests that these extreme rates are not systematically associated with RML.

To examine the effects of RML on students' time use, I construct time-use measures for individuals between 2010 and 2023 using ATUS data, following the activity classification in Chu and Gershenson (2018). I refine this classification by separating education-related time into *Study at School* and *Study at Home*. *Study at School* includes attending class and participation in non-sport extracurricular school activities, while *Study at Home* corresponds to time spent on research or homework. I also revise the classification of leisure time by decomposing it into positive leisure time and passive leisure time. Appendix A3 describes the detailed classification, and Table A11 lists the corresponding activity codes.

I use the 2011–2023 waves of the PSID to construct individual-level measures of GPA, substance use, and other health behavior, including drinking alcohol and smoking cigarettes.[12] Because the PSID is subject to substantial item nonresponse and panel attrition, with many respondents not observed in every wave, the analysis relies on an unbalanced panel. GPA is measured using respondents' self-reported grade point average at their current college. The GPA analysis is therefore restricted to respondents ages 18–24 who are enrolled in college at the time of the TAS interview. I also use the PSID measure *In College*, an indicator for current college enrollment, to examine whether RML changes enrollment among respondents ages 18–24. This variable is available in the study's extracted data only for 2011, 2013, and 2015, so the enrollment analysis covers a shorter period than the GPA analysis. For substance use, I focus on marijuana and other substances and construct measures of both ever use and use frequency. The available observations on heroin, hallucinogens, inhalants, and pain relievers provide insufficient support for the analysis, so this paper does not examine outcomes for these substances. Appendix A1 gives the frequency-coding rules.

The time-use, substance-use, and other health-behavior analyses include individuals ages 18–24 regardless of current college enrollment, a range over which college enrollment is common. The In College analysis uses the same age restriction and includes both enrolled and non-enrolled respondents.[13] The GPA estimates, by contrast, are conditional on current enrollment and may be affected by changes in the composition of enrolled students.

Table A12 reports placebo tests for changes in disclosure after RML. Appendix A4 describes the two sources of reporting bias and the checks used to address them. For every substance-use measure, none of the resulting p-values

[12]Specifically, the 2011, 2013, ..., 2023 waves.

[13]Bailey and Dynarski (2011) classify a "college completer" as someone who, at age 25, has completed sixteen years of schooling or holds at least a BA when measuring educational inequality.

falls below 0.1, suggesting that this type of reporting bias is not statistically significant and is therefore unlikely to meaningfully affect the main results.

Table A1 reports the policy-timing regressions described in Appendix A2. Taken together, the results suggest that, prior to legalization, the main outcomes are not systematically related to legalization timing, providing supportive evidence for the identifying assumption.

4 RML and Graduation Rate

4.1 Identification

The primary objective of this paper is to identify the causal effects of RML on higher-education outcomes. I focus on bachelor's degree completion as a central measure of higher-education attainment. A major threat to identification is that RML may change the composition of enrolled students by affecting college application and enrollment decisions, thereby confounding estimates of post-enrollment outcomes. If college entry responds to RML, estimates based on the full population of potential students may conflate changes in who enrolls with changes in outcomes among those already enrolled, leading to bias from endogenous selection.

To address this concern, I restrict the analysis to students who were already enrolled at the time of legalization. By conditioning on pre-policy enrollment, this design largely eliminates the enrollment margin as a source of variation and mitigates endogeneity arising from RML-induced selection, allowing for a cleaner estimate of RML's effect on educational outcomes such as bachelor's degree completion. To examine the effects of RML on graduation outcomes, I estimate the following difference-in-differences model for institution i:

$$Y_{ic} = \beta M_{ic} + \nu_i + \nu_c + \epsilon_{ic} \tag{1}$$

Let Y_{ic} denote the graduation outcome for institution i and cohort c, where cohort timing is indexed in academic years since first enrollment. The variable M_{ic} is an indicator equal to one if cohort c in institution i was exposed to RML in its state before the relevant graduation outcome was measured, and zero otherwise. The coefficient β captures the average treatment effect of RML. In addition, ν_i denotes institution fixed effects, and ν_c denotes cohort fixed effects. Standard errors are clustered at the state level. By including institution and cohort fixed effects, the specification controls for time-invariant differences across institutions and common shocks across cohorts.

Appendix A2 gives the graduation event windows and reference cohorts.

Recent work shows that in staggered DiD designs, the conventional two-way fixed effects (TWFE) estimator can place nontrivial weight on biased "forbidden comparisons," underscoring the need for estimators that are robust to treatment-effect heterogeneity (Goodman-Bacon 2021). In the present setting, the Goodman–Bacon decomposition indicates that about 55% of the TWFE estimate is driven by treated-versus-treated timing comparisons, while only about 45% comes from comparisons between treated and never-treated states. Under heterogeneous treatment effects, the former comparisons are the problematic "forbidden" components. Consistent with this decomposition, the overall TWFE estimate is smaller than the treated-versus-never-treated component, suggesting attenuation from timing-based comparisons. I therefore rely on estimators designed for staggered adoption under heterogeneous treatment effects, such as group-time ATT and event-study estimators. These results highlight the limitations of the conventional TWFE estimator in this setting and reinforce the importance of using alternative methods that explicitly accommodate treatment-effect heterogeneity.

To estimate the causal effect of marijuana legalization while addressing the limitations of conventional staggered DiD regressions, I employ the estimator developed by Callaway and Sant'Anna (2021). Recent methodological contributions have emphasized that treatment effects may vary across cohorts and over time, rendering traditional two-way fixed effects estimators difficult to interpret in staggered adoption settings (De Chaisemartin and d'Haultfoeuille (2020); Sun and Abraham (2021); Wooldridge (2021); Borusyak, Jaravel and Spiess (2024)). The Callaway and Sant'Anna (2021) framework is particularly attractive in this context because it constructs valid comparisons using never-treated or not-yet-treated units, yields interpretable group-time average treatment effects, and permits flexible aggregation across treatment cohorts and calendar periods.

Consistent with Callaway and Sant'Anna (2021) and the broader staggered-adoption DiD literature, which is largely formulated for a single-treatment setting (Roth et al. 2023), I model RML as the sole policy intervention of interest. I therefore do not include MML as a contemporaneous control, since the mechanisms through which medical and recreational legalization affect graduation outcomes are likely to overlap, making it difficult to separately identify their effects within this framework. Instead, to minimize contamination from overlapping policy exposure, I restrict the sample to states whose MML treatment year occurred on or before 2010.[14] Under this restriction, the analysis sample

[14]To provide a more comprehensive analysis, samples based on alternative cutoff years are examined in the robustness checks. The definition of the treatment year is provided later in this section.

consists of 15 states that adopted RML before 2023 and 1 state that did not adopt RML during the study period.[15]

Given the limited number of state clusters in the analysis sample, conventional cluster-robust asymptotic inference may perform poorly. I therefore cluster standard errors at the state level and rely on the wild bootstrap procedure implemented in the Callaway and Sant'Anna (2021) framework for inference.[16] This approach is especially useful when the number of clusters, or treated clusters, is limited.

In a DiD setting, identification rests on the parallel trends assumption: absent treatment, treated and comparison units would have evolved along the same counterfactual trajectory of the outcome over time. Following Callaway and Sant'Anna (2021), I estimate group–time average treatment effects, $ATT(g,t)$, which are defined for each adoption cohort g and calendar period t using never- or not-yet-treated units as valid comparisons. Their framework also delivers dynamic event-study effects by indexing periods relative to each unit's first treatment date. This structure permits both (i) recovery of ATT's for all cohorts and periods and (ii) transparent assessment of the parallel trends assumption via pre-treatment "lead" coefficients: under parallel trends, the estimated pre-treatment effects should be close to zero and statistically indistinguishable from zero across cohorts.

I recode state treatment years so that enactments occurring on or after September 1 are assigned to the subsequent calendar year. Legalization months vary across states, and because most degrees are awarded before September-and NCES (IPEDS) records completions as of August 31-enactments before September can plausibly affect the current graduating cohort, whereas enactments in September-December can only affect the next cohort. This coding rule ensures that RML can affect admissions and enrollments only for students whose entering cohort falls in the treatment year or later, and not for students who enrolled before the treatment year. As a result, I can examine the impact of RML on cohorts who are in academic years 1-4 at the time of legalization without contaminating the estimates with composition changes in the pool of entering students.

4.2 Graduation Rate Results

Figure 2 panels A-C report event-study estimates of the effect of RML on BA graduation rates for students who were in their first through fourth academic years at the time their state legalized recreational marijuana. In each specification, the reference cohort is defined as students who were in their fifth, sixth,

[15]Table 2 reports the list of states included in the sample.

[16]Specifically, I use the `wboot` option.

and seventh academic years at the time of legalization when the outcome is BA completion within 4, 5, and 6 years, respectively. Panel D presents the corresponding estimate for graduation within 3 years at two-year institutions, where the reference cohort consists of students who were in their fourth academic year at the time of legalization. The event-study results suggest that the implementation of RML is associated with a decline in graduation rates, with the largest effect observed for the cohort in its first academic year at the time of implementation. The main reason is that these students are at the margin of exposure: recreational marijuana becomes legal almost immediately after they enroll, so they are effectively exposed to RML throughout nearly their entire time in college. In other words, this cohort is closer to being a fully treated group.

Across all event-study specifications, the pre-treatment estimates are statistically insignificant and centered around zero, lending strong support to the parallel-trends assumption. Moreover, the absence of any systematic pre-treatment pattern suggests that treated and reference cohorts followed similar graduation trajectories prior to legalization, thereby strengthening the credibility of the identification strategy and supporting a causal interpretation of the post-treatment effects.

Table 3 reports the average treatment effects on the treated (ATT) of RML estimated using the Callaway and Sant'Anna (2021) difference-in-differences estimator for cohorts that entered college between 2006 and 2017.[17] Columns (1), (2), and (3) present graduation rate for 4 year institution. Column (1) shows that RML significantly reduces the probability that enrolled students complete a BA within four years by 1.6 percentage points, which corresponds to a 3.8% decline relative to the pre-treatment mean of 0.419. Columns (2) and (3) indicate that RML significantly lowers the probability of completing a BA within five and six years by 1.1 and 1.8 percentage points, respectively, representing declines of 2.0% and 3.1% relative to the pre-treatment means of 0.548 and 0.584. The estimated effect on five-year graduation rates (1.1 pp) is meaningfully smaller than the corresponding effect on four-year graduation rates (1.6 pp). This attenuation suggests that some students induced not to graduate within four years eventually complete their degree within five years, pointing to a delay effect of RML on BA attainment. However, the effect on six-year graduation rates remains statistically significant and negative, implying that the impact of RML is not solely driven by delayed graduation. Instead, the results are consistent with

[17]Specifically, I implement the estimation in Stata using the *simple* option of the *csdid* command.

both a timing effect and a persistent reduction in ultimate degree completion, whereby some students may be permanently deterred from obtaining a BA.

Column (4) reports the estimated effect of RML on graduation rates at two-year institutions. The results show that RML significantly lowers the probability that enrolled students graduate within three years by 2.0 percentage points, which represents a 6.0% decline relative to the pre-treatment mean of 0.336. This magnitude is substantially larger than the estimated effect on BA completion, suggesting that students in non-BA programs may be more vulnerable to the adverse effects of RML. Moreover, because these students do not complete a two-year program even within three years, the estimates are consistent with a persistent—and possibly permanent—reduction in degree completion.

An important question is whether the effect of RML on graduation rates varies with institution cohort size. To examine this, I classify institutions according to the largest cohort size they have ever recorded over the sample period and estimate the effect of RML separately across these groups. Figure A1 presents the corresponding estimates by institution cohort size. The results indicate that the negative effect of RML on graduation outcomes is more pronounced among institutions with smaller cohort sizes than among those with larger cohort sizes. For BA completion within 4 and 5 years, the estimated effect gradually attenuates as cohort size increases, although it remains statistically significant throughout. By contrast, for BA completion within 6 years and for graduation rates at two-year institutions, the estimated effects are largely stable across cohort-size groups, with the exception of institutions with a cohort size of 500, where the negative effect appears slightly larger.

To complement the institution-level analysis, Figure A2 reports the corresponding estimates by institution cohort size weighted by cohort size, thereby capturing the average effect of RML at the student level rather than the average effect across institutions. The weighted estimates show a similar but more nuanced pattern. For BA completion within 4 and 5 years, the average effect of RML on individual students remains statistically significant among institutions with smaller cohort sizes, but becomes statistically insignificant among institutions with larger cohort sizes. By contrast, for BA completion within 6 years and for graduation rates at two-year institutions, the estimated effects remain broadly stable across cohort-size groups. This pattern suggests that, in larger-cohort institutions, the average student-level effect of RML is less reflected in short-run delays in graduation and more likely to take the form of persistent, potentially permanent, reductions in degree completion.

Students may be affected differently by RML depending on the type of institution they attend. Although the data do not allow for a detailed identification

of individual student characteristics, institutional type may partially capture differences in student composition through sorting. Table A13 reports estimates of the effect of RML on BA graduation rates by institution type. Panel A presents separate estimates for private and public institutions.[18] A striking pattern is that the negative effect of RML on BA graduation rates is concentrated almost entirely among private institutions, while the corresponding estimates for public institutions are close to zero and statistically insignificant. Panel B reports heterogeneity by Carnegie Classification.[19]

For BA completion within 4 and 5 years, the adverse effect of RML is concentrated primarily among institutions whose highest degree awarded is at the baccalaureate/associate level. By contrast, for BA completion within 6 years, the negative effect is concentrated mainly among doctoral- and master's-granting institutions.

RML-induced declines in BA attainment are likely to have meaningful long-run consequences in the labor market. A broad literature shows that completing a bachelor's degree is associated with substantially higher earnings than not completing one. For example, Carnevale, Rose and Cheah (2011) document that, over the life cycle, workers with a bachelor's degree earn about 84% more than those with only a high school diploma. In addition, Deming (2023) show that the college wage premium is not constant over the life cycle, but instead rises from about 27% at age 25 to about 60% at age 55. These patterns imply that even a modest reduction in BA completion can translate into sizable cumulative earnings losses over time. More importantly, evidence from papers with stronger causal identification suggests that these losses are not merely descriptive correlations. Using university dismissal thresholds in a regression discontinuity design, Ost, Pan and Webber (2018) show that students just below the cutoff are about 10 percentage points less likely to obtain a BA and experience about a 5.6% reduction in mid-run weekly earnings relative to similar students just above the cutoff. Likewise, using admissions thresholds for academically marginal students, Zimmerman (2014) find that marginal admission to a four-year public university raises earnings by about 22% between 8 and 14 years after high school completion. These results are especially relevant for the present study, because RML is unlikely to affect all students equally; instead, it may disproportionately push academically marginal students off the college completion margin. Therefore, if RML reduces BA graduation rates, the resulting educational losses should also be expected to appear in the form of lower long-run earnings in the labor market.

[18] Because the number of private two-year institutions in the sample is too small, this analysis is restricted to four-year institutions.

[19] Appendix A1 gives the classification rules.

4.3 Heterogeneity

This section examines the heterogeneous effects of RML on graduation rates by gender and race. Table A14 reports the ATT estimates from the event-study specifications. Columns (1)–(3) present estimates for four-year institutions. The negative effects of RML on graduation within 4, 5, and 6 years are consistently larger in magnitude and statistically significant for males, whereas the corresponding estimates for females are smaller, despite females having higher pre-treatment graduation rates than males.

The effects of RML also vary across racial groups in four-year institutions. For graduation within 4 years, the negative effect is statistically significant only for White students. For graduation within 5 and 6 years, the estimates remain significant for White students and also become significant for Hispanic and Asian students. By contrast, the estimates for Black students are not statistically significant in any specification.

Column (4) presents estimates for two-year institutions. RML has statistically significant negative effects for both female and male students, although the magnitude of the effect is larger for females than for males. Notably, once the sample is divided by race, none of the race-specific estimates is statistically significant. Although RML has a statistically significant effect in the overall sample, the subgroup analyses by race do not yield statistically significant estimates for White, Black, Hispanic, or Asian students individually. One possible explanation is that splitting the sample by race substantially reduces precision and statistical power, making it more difficult to detect effects within each subgroup even if the underlying impacts are similar in direction.

5 The Effects of RML on GPA and Time Use

5.1 Identification

Examining the effects of RML on the intermediate academic outcomes of college students is also an important part of the analysis. In this section, I focus on how RML affects college students' GPA and their allocation of time in daily life. These outcomes are not only educational outcomes in their own right, but also potential indirect mechanisms through which RML may ultimately affect graduation rates.

I restrict the GPA and time-use analyses to individuals ages 18–24, when college attendance is most common.[20] GPA is observed for respondents currently enrolled in college, whereas the time-use analysis includes respondents regardless

[20]Both outcomes are estimated using the Callaway and Sant'Anna (2021) method.

of enrollment status. I also estimate the effect on In College among respondents in this age range using the available 2011–2015 waves to assess whether enrollment responds to RML. For the effect of RML on GPA, I then estimate the following DiD specification:

$$Y_{it} = \gamma M_{it} + \nu_i + \nu_t + \epsilon_{it} \tag{2}$$

Y_{it} denotes the outcome of interest for individual i in period t, including self-reported GPA and current college enrollment. The period index t corresponds to biennial survey years: 2011–2023 for GPA and 2011–2015 for In College.[21] This formulation allows the same empirical framework to be applied to multiple student-level outcomes while maintaining a consistent identification strategy across specifications. The regression includes both individual fixed effects and period fixed effects, which absorb time-invariant individual heterogeneity and common shocks across periods, respectively.

For the GPA analysis, the dependent variable is self-reported GPA, measured on a 4.0 scale. The primary parameter of interest is γ, which captures the change in GPA associated with RML exposure under the DiD design. Interpreted in this way, γ reflects how RML affects academic performance among currently enrolled students ages 18–24, conditional on the controls and fixed effects included in the model.

In addition, following Chu and Gershenson (2018) and related studies on the effects of MML on students' time use, I extend this line of research to examine the impact of RML on students' time allocation. In particular, for individual i in state s, I estimate the following difference-in-differences specification:

$$Y_{ist} = \delta M_{ist} + C_i + \nu_s + \nu_t + \epsilon_{ist} \tag{3}$$

Y_{ist} denotes the outcome of interest for individual i in period t and state s, including self-reported daily time spent studying at home, studying at school, and engaging in leisure activities during 2010-2023. Ci is a vector of controls that captures the timing of respondent i's diary, including diary-month fixed effects, day-of-week fixed effects, and a dummy variable indicating whether the diary was collected on a holiday, and the age of individual i. The regression further includes state fixed effects and time fixed effects.

[21] Each period corresponds to two years. In later sections, Equation (2) is also used to estimate the effect of RML on individual drug use outcomes.

5.2 Academic Performance and College Attendance

RML significantly reduces GPA among currently enrolled college students ages 18–24. The estimated decline is 0.350 points, or 10.5% of the estimation-sample pre-treatment mean of 3.33 (Table 4, column (1)). Figure 3, Panel A, shows negative post-treatment estimates that become larger in magnitude over the observed horizon. The displayed pre-treatment estimate is statistically insignificant.[22]

Because GPA is observed only for students who are currently enrolled, changes in who remains in college could affect its interpretation. I therefore also examine *In College*, an indicator of current college attendance among all respondents ages 18–24. This measure captures participation beyond initial entry: attendance also depends on continued study, dropout or temporary withdrawal, and degree completion. The estimated effect is an increase of 2.0 percentage points and is statistically insignificant (Table 4, column (2)). Panel B of Figure 3 reports the corresponding event-study estimates.

The attendance result offers only a limited check on selection into the GPA sample: the available data cover 2011–2015, and there is an earlier pre-treatment difference.[23] The decline in GPA is consistent with weaker academic progress among students who remain enrolled, a possible channel linking RML to lower degree completion. These results do not establish how much of the graduation effect operates through academic performance.

5.3 Time Use Outcomes

Figure 4 presents the event-study estimates of the impact of RML on students' time use. The pre-treatment coefficients are close to zero and statistically insignificant across all specifications, which is consistent with the parallel-trends assumption. Following legalization, RML does not generate immediately large or consistently significant effects on study time or leisure time. Nevertheless, the estimates reveal a clear negative post-treatment pattern for study time at home and a corresponding positive pattern for passive leisure time.

Table 5 reports the ATT estimates for time use. The estimated effects of RML on study time at school and positive leisure time are both statistically insignificant, although the point estimate is positive in each case. By contrast, RML reduces study time at home by 44.43 minutes, an effect whose magnitude exceeds the pre-treatment mean of 37.39 minutes. At the same time, RML

[22] Appendix A2 discusses the precision of the dynamic and aggregate estimates.

[23] The attendance estimate at event time -3 is negative and statistically significant. It falls outside the common -2 to 1 window displayed in Figure 3. Current attendance does not distinguish dropout from other entry and exit decisions, and an insignificant average effect does not exclude changes in the composition of enrolled students.

increases passive leisure time by 54.17 minutes, which corresponds to 28.3% of its pre-treatment mean of 191.46 minutes. As with the GPA results, the overall ATT pools information across all post-treatment periods, which improves statistical power and allows for a more precise estimate of the average treatment effect.

These findings are not only consistent with Chu and Gershenson (2018) on the effects of MML on students' time use, but also extend their analysis to the context of RML. Chu and Gershenson (2018) report that MML reduces the time students devote to educational activities by 20 percent and increases the time they spend on leisure activities by 20 percent. Using a more detailed decomposition of both study time and leisure time, my results suggest that the effects of RML on students' educational and leisure time allocation are larger in magnitude than those of MML. More importantly, the time-use results provide additional insight into another indirect mechanism through which RML may reduce graduation rates: students appear to devote less time to studying, particularly to self-directed study at home, which may in turn undermine academic progress and degree completion. Taken together, the evidence on GPA and time use is consistent with the graduation-rate results and further suggests that RML has detrimental effects on higher-education outcomes.

6 Mechanism: Effects of RML on Substance Use and Related Health Behaviors

6.1 Identification

Discussions of RML naturally center on marijuana use itself. As the most immediate mechanism behind the main findings of this paper, this section first examines the impact of RML on marijuana consumption. In addition, it explores another important channel through which RML may shape higher-education outcomes: its effect on other substance use, alcohol consumption, and cigarette smoking.

A key mechanism through which marijuana affect educational attainment is its negative impact on cognitive function and learning capacity. A growing body of research highlights the detrimental effects of marijuana use on both cognitive functioning and broader life outcomes. Regular and cumulative use has been shown to impair key cognitive abilities—including memory, executive functioning, emotional regulation, and processing speed—with particularly strong effects on verbal memory (Crean, Crane and Mason (2011); Auer et al. (2016)). Adolescents appear especially vulnerable, exhibiting significant declines in IQ and verbal learning relative to their non-using peers, suggesting lasting cognitive

consequences when use occurs during critical developmental periods (Lorenzetti, Hoch and Hall (2020)). Moreover, frequent marijuana use in adolescence is consistently associated with adverse psychosocial outcomes in adulthood, including diminished life satisfaction, and increased risk of mental health disorders such as depression, anxiety, and psychosis (Hall (2015)).

The use of other illicit drugs may also impair individuals' learning ability. A large medical and neuropsychological literature shows that long-term use of cocaine, amphetamines, and opioids is associated with deficits in attention, learning and memory, verbal memory, and broader cognitive functioning (Jovanovski, Erb and Zakzanis 2005; Potvin et al. 2014; Frazer, Richards and Keith 2018; Mahoney III 2019; Ersche and Sahakian 2007; Janke van Holst and Schilt 2011). Hallucinogens, in turn, are known to produce substantial changes in perception, mood, and a range of cognitive processes (Nichols 2004). In addition, the use of appearance drugs appears to be associated with several psychopathological disorders, although their prevalence remains unclear (Zaami et al. 2021). More importantly, the adverse consequences of these substances may be greater than those associated with marijuana use. Lundqvist (2005) suggest that although cannabis use impairs attention, memory, and executive functioning, the cognitive deficits linked to stimulant and heroin abuse are often broader and more severe. Similarly, Janke van Holst and Schilt (2011) show that, relative to cannabis, the use of cocaine, (meth)amphetamines, ecstasy, and opiates is more consistently associated with persistent deficits in executive functioning, particularly in inhibition. At the same time, alcohol consumption and smoking may also adversely affect higher educational outcomes. Prior research suggests that alcohol and tobacco use may undermine educational attainment through several channels, including reduced study time, lower academic motivation and engagement, greater absenteeism, and impaired cognition and self-regulation (Alqahtani et al. 2023; Balsa, Giuliano and French 2011; El Ansari, Stock and Mills 2013; Janke van Holst and Schilt 2011; Pokhrel et al. 2013).

Marijuana use is closely related to the use of other illicit drugs and to a range of associated health behaviors. Early evidence suggests that marijuana may be complementary to the consumption of other substances. Grossman and Chaloupka (1998) provide evidence that cocaine and marijuana are complements in consumption, while Pacula et al. (2001) show that both beer and cocaine may be complementary to marijuana use. There is also evidence that marijuana use increases the probability that individuals with no prior cocaine use initiate cocaine consumption (DeSimone 1998). More broadly, Melberg, Jones and Bretteville-Jensen (2010) find that cannabis may serve as a gateway to hard drugs, although this effect appears to be concentrated among a small subgroup

of high-risk youths and is weak and statistically insignificant for most youths. Likewise, Hall and Lynskey (2005) note that marijuana use typically precedes the use of other illicit drugs, and that earlier initiation and more frequent marijuana use are associated with a higher probability of subsequent use of other illicit substances. Figures A3 and A4 present the relationships between marijuana use and other substance use, alcohol consumption, and cigarette smoking. Overall, marijuana use is positively associated with these behaviors. The medication measures may include prescribed use, and these descriptive relationships do not distinguish medical from nonmedical use.

I estimate the effects of RML on marijuana and other substance use, alcohol consumption, and cigarette smoking using individual-level data from the PSID. The empirical specification follows equation 2, with all variables defined in an analogous manner. The outcome variable Y_{it} captures either an indicator for lifetime marijuana or cocaine use or a measure of use frequency. Consistent with the GPA analysis, t indexes the biennial survey waves from 2011 to 2023, and the estimation sample is restricted to individuals aged 18 to 24. Because some substance use measures are missing for a subset of the PSID sample, the event-study analysis for substance use includes only two lead and two lag bin, corresponding to the four years before and the four years after the treatment year, respectively. For smoking and drinking, I use a wider event window, with the pre-treatment period extending to six years before treatment and the post-treatment period extending to four years after treatment.

6.2 Substance Use, Drink and Smoke Results

Figure 5 reports the event-study estimates of the effects of RML on marijuana use among individuals aged 18–24. In Panel A, the coefficient for *Pre 2* in marijuana ever use is significantly negative, although the overall pre-treatment pattern appears relatively flat. The post-treatment coefficients are positive, but statistically insignificant. By contrast, in Panels B, C, and D, which examine marijuana use frequency, the pre-treatment coefficients are generally flat and statistically insignificant, while the post-treatment coefficients display an upward pattern, even though some individual estimates are not statistically significant. Table 6 reports the corresponding ATT estimates for marijuana use. The estimated effect of RML on marijuana ever use is positive, but not statistically significant. In contrast, RML significantly increases the frequency of marijuana use in the past 30 days, past 12 months, and over the lifetime by 1.26, 2.54, and 2.75 score points, respectively. These estimates correspond to increases of 29%, 36%, and 26% relative to the estimation-sample pre-treatment means of 4.37, 6.97, and 10.59.

Table A15 reports the ATT estimates of the effects of RML on other drug use. Overall, RML has positive and statistically significant effects on cocaine use and on the frequency scores for amphetamines, barbiturates, and tranquilizers (Group A). For Group B (diet pills and steroids), both ever use and the lifetime frequency score increase significantly. Specifically, for cocaine, RML significantly increases ever use by 4.4 percentage points, corresponding to a 52% increase relative to the estimation-sample pre-treatment mean of 0.085. RML also increases the frequency of cocaine use in the past 30 days, past 12 months, and over the lifetime by 0.072, 0.322, and 0.999 score points, respectively, which correspond to increases of 151%, 116%, and 125% relative to the estimation-sample pre-treatment means of 0.048, 0.277, and 0.802. For Group A, RML has no statistically significant effect on ever use, but it significantly increases frequency scores in the past 30 days, past 12 months, and over the lifetime by 0.985, 1.92, and 1.84 points, respectively, corresponding to increases of 106%, 81%, and 36% relative to the estimation-sample pre-treatment means of 0.933, 2.38, and 5.18. Finally, for Group B, RML significantly increases ever use by 8.5 percentage points, which is equivalent to a 70% increase relative to the estimation-sample pre-treatment mean of 0.121. The lifetime frequency score also increases significantly, by 1.45 points, or 121% of its pre-treatment mean of 1.20. Consistent with the ATT estimates reported in Table A15, Figures A5, A6, and A7 present the corresponding event-study estimates for the various measures of other drug use.

Figure A8 reports the event-study estimates of the effects of RML on drinking and smoking. Across all outcomes, the pre-treatment coefficients are broadly flat and statistically insignificant, providing support for the parallel-trends assumption. For ever drinking, the post-treatment coefficients are positive and display an upward trend over time, with the estimates in periods 0 and 1 statistically significant. For drink quantity, the post-treatment coefficients have mixed signs, and none is statistically significant. For ever smoking, the post-treatment coefficients also show an upward trend, but no individual post-treatment estimate is statistically significant. Table A16 presents the corresponding ATT estimates. The results indicate that RML increases the probability of ever drinking by 21.3 percentage points, which corresponds to a 34% increase relative to the estimation-sample pre-treatment mean of 0.632. In contrast, RML has no statistically significant effect on drink quantity. At the same time, RML increases the probability of ever smoking by 7.2 percentage points, equivalent to a 33% increase relative to the estimation-sample pre-treatment mean of 0.215.

The results above indicate that RML increases marijuana-use frequency scores, while its effect on reported ever use is not statistically significant.

Because the frequency measures include nonusers, they can reflect both participation and changes in use among existing users. The estimates do not isolate an intensive-margin effect. More generally, the findings suggest increases in marijuana and other substance use, with potentially adverse consequences for educational and health outcomes.

My finding that RML increases marijuana use among young adults is consistent with a substantial body of prior research (Bae and Kerr 2020; Barker and Moreno 2021; Gali et al. 2021; Miller, Rosenman and Cowan 2017; Stormshak et al. 2019; Subbaraman and Kerr 2020). The increases in use-frequency scores are also consistent with the evidence in Papich (2022). Moreover, my finding that RML increases other substance use is consistent with (Mathur and Ruhm 2023; Liu, Xie and Chen 2025; Meinhofer, Rubli and Cunningham 2026), while the result that RML increases the likelihood of ever drinking alcohol accords with the evidence in Paschall, García-Ramírez and Grube (2022). At the same time, other strands of the literature suggest that RML may reduce alcohol and cigarette consumption (Dave et al. 2023; De and Sun 2025), while several studies find little or no effect of marijuana legalization on the use of harder substances(Sabia et al. 2021; Mennis, Stahler and Mason 2021; Anderson and Rees 2023). More generally, there is limited evidence that RML increases the use of hard drugs (Sabia et al. 2021).

7 Robustness Checks

7.1 Inconsistent Self-Reports

Table A17 presents robustness checks on the main results by accounting for the second type of reporting bias discussed in Section 3 when measuring ever use of substances, drinking, and smoking. The checks impose consistent ever-use histories or exclude respondents with inconsistent reports; Appendix A4 gives the coding rules. In Column (1), the estimated effect of RML on marijuana ever use is positive and statistically significant. Compared with the main results, accounting for the second type of reporting bias uncovers an extensive-margin effect of RML on marijuana use. Therefore, the possibility that RML leads some previously non-using individuals to initiate marijuana use cannot be completely ruled out. For drink quantity, the estimated effect after excluding respondents with reversals in reported drinking status is -0.084 and is statistically insignificant, as in the baseline specification. These estimates do not establish a reduction in drinking intensity or heavy drinking.

7.2 Alternative MML Cutoffs

The main results are based on a sample constructed using 2010 as the cutoff year, which restricts the analysis to states in which MML treatment occurred before 2010. This restriction allows the sample to include a broader set of eligible states while maintaining the identifying assumption that MML does not contaminate the pre-treatment period. Changing the cutoff year results in modest variation in the set of states included in the sample. Tables A4–A10 show that the estimated effects of RML on the main outcomes are robust to these alternative sample definitions.

In Table A4, each column presents the estimated effect of RML on graduation rates based on a different cutoff year. Appendix A5 describes how the cohort range changes with the cutoff year. As the cutoff year moves forward, more states and institutions are included in the sample, leading to a larger estimation sample. At the same time, the estimated effects of RML on all graduation outcomes become smaller in magnitude. For BA completion within 4 and within 5 years, the estimated effects are statistically insignificant when the cutoff year is set between 2005 and 2009. In contrast, the effect of RML on BA completion within 6 years, as well as its effect on graduation rates at 2-year institutions, remains statistically significant throughout.

Tables A5–A10 report the corresponding estimates for the remaining outcomes under alternative cutoff years. Panel B of Table A5 reports the effects of RML on time use under alternative cutoff years. The estimated effect of RML on passive leisure is statistically insignificant for most cutoff years, whereas the estimated effect on positive leisure is positive across specifications and statistically significant in some cases. These results complement the main findings on time use. For the other outcomes, the estimates obtained using relatively later cutoff years are broadly similar to the baseline results based on the 2010 cutoff year, while covering a larger set of states and observations. For earlier cutoff years, some estimates lose statistical significance, but the majority continue to display the same sign as the main results. Taken together, these patterns support the robustness of the empirical strategy. While the approach requires sacrificing some observations, it substantially improves the ability to distinguish the effects of RML from those of MML.

8 Discussion

The substance-use estimates offer evidence on possible mechanisms. Among individuals ages 18–24, RML increases marijuana-use frequency scores by 29%,

36%, and 26% for the past 30 days, past 12 months, and lifetime measures, respectively. Both ever use and frequency scores rise for cocaine. I also find higher frequency scores for Group A and increases in ever use and lifetime frequency scores for Group B (diet pills and steroids). RML increases the probability of ever drinking alcohol and smoking cigarettes, while having no statistically significant effect on drink quantity. The drinking finding is consistent with Paschall, García-Ramírez and Grube (2022); a complementary relationship between marijuana and alcohol may help explain it, as Wen, Hockenberry and Cummings (2015) suggest for MML. These policy effects on substance use do not identify how much of the educational effect operates through any particular substance.

Appendix A6 reports complementary evidence on MML and BA completion. MML lowers four-year completion and raises five-year completion, while the six-year estimate is statistically insignificant. This pattern is consistent with delayed graduation, although longer-run effects remain uncertain. The reduction in on-time completion is consistent with the adverse educational outcomes documented by Plunk et al. (2016), who find lower college degree completion among those exposed to MML during adolescence. Chu (2015) provides related evidence on a potential behavioral channel: MML increases proxies for marijuana consumption, but there is no evidence of increased cocaine or heroin use.

Compared with MML, the possible mechanisms linking RML to educational outcomes are broader and potentially more consequential. First, RML expands marijuana access to a larger share of the population. Relative to MML, RML is associated with larger increases in the intensity of marijuana use, with consumption becoming more frequent and more persistent over time. This greater exposure may amplify the long-run harms of use. Second, beyond increasing marijuana use, RML also leads to greater reported use of other substances, whose effects on educational attainment may be even more detrimental. This distinction is important because such a pattern has not been documented for medical marijuana legalization. Prior work finds no evidence that medical marijuana laws increase cocaine or heroin use (Chu 2015), whereas Mathur and Ruhm (2023); Liu, Xie and Chen (2025); Meinhofer, Rubli and Cunningham (2026) provide evidence that RML significantly increases the use of other illicit drugs. Third, RML may more strongly alter social norms and perceptions of risk. Educational outcomes may be affected not only through greater access to marijuana, but also through changes in students' beliefs: they may perceive marijuana use as more socially acceptable, less harmful, and less costly in social terms. These changes may in turn affect study effort, self-control, and time allocation. Consistent with this interpretation, existing evidence shows that RML increases marijuana use

among young people and strengthens the relationship between low perceived risk and marijuana use after legalization (Mennis, McKeon and Stahler 2023).

9 Conclusion

This paper examines the relationship between marijuana legalization and educational attainment. Exploiting the staggered rollout of RML across states, I estimate its causal effects on higher-education attainment. I find that RML significantly reduces the probability that full-time, first-time students earn a bachelor's degree. The dynamic estimates suggest that RML operates not only by delaying BA completion—shifting some students from on-time graduation to completion in five or six years—but also by leaving a subset of students without a degree even six years after initial enrollment, indicating a persistent long-run negative effect on educational attainment. In addition, RML significantly lowers the three-year graduation rate at 2-year institutions, where the estimated effect appears to reflect a more permanent reduction in completion rather than a temporary delay.

In addition, the paper examines the effects of RML on two education-related outcomes, GPA among currently enrolled college students and students' time allocation. The negative effects of RML on these outcomes are consistent with the broader pattern of adverse effects on educational attainment. Examining these margins provides further insight into the indirect channels through which RML may influence educational attainment. The shorter 2011–2015 series shows no statistically significant effect on current college enrollment among 18–24-year-olds, although this does not rule out selection in the GPA sample.

I also provide evidence on two possible mechanisms through which RML may affect educational outcomes. First, RML increases marijuana-use frequency scores among college-age individuals. Second, RML increases both initiation into other drugs use, drink and smoke. The baseline effect on marijuana ever use is statistically insignificant, while the frequency-score effects are significant; because these scores include nonusers, they do not isolate an intensive margin. By contrast, the effects of RML on other drugs are concentrated mainly in increased use of cocaine, higher Group A frequency scores, and increased ever use and lifetime scores for Group B (diet pills and steroids). For drinking and smoking, the effects of RML also appear to arise primarily along the extensive margin. These findings have important policy relevance. In states where RML is already in place, they suggest the need for closer attention to the regulation of marijuana use intensity, since the educational harms appear to be linked less to occasional exposure than to frequent and persistent use. They also highlight

the importance of monitoring and regulating other drugs, and of strengthening enforcement against black-market activity. For states that have not yet legalized recreational marijuana, the results suggest that policymakers should weigh these potential educational costs carefully, possibly alongside stricter age thresholds or other measures designed to reduce exposure among young adults.

Overall, this paper provides the first comprehensive analysis of the educational consequences of recreational marijuana legalization. The findings carry important implications for the design of marijuana policies and establish a strong foundation for future work in this area.

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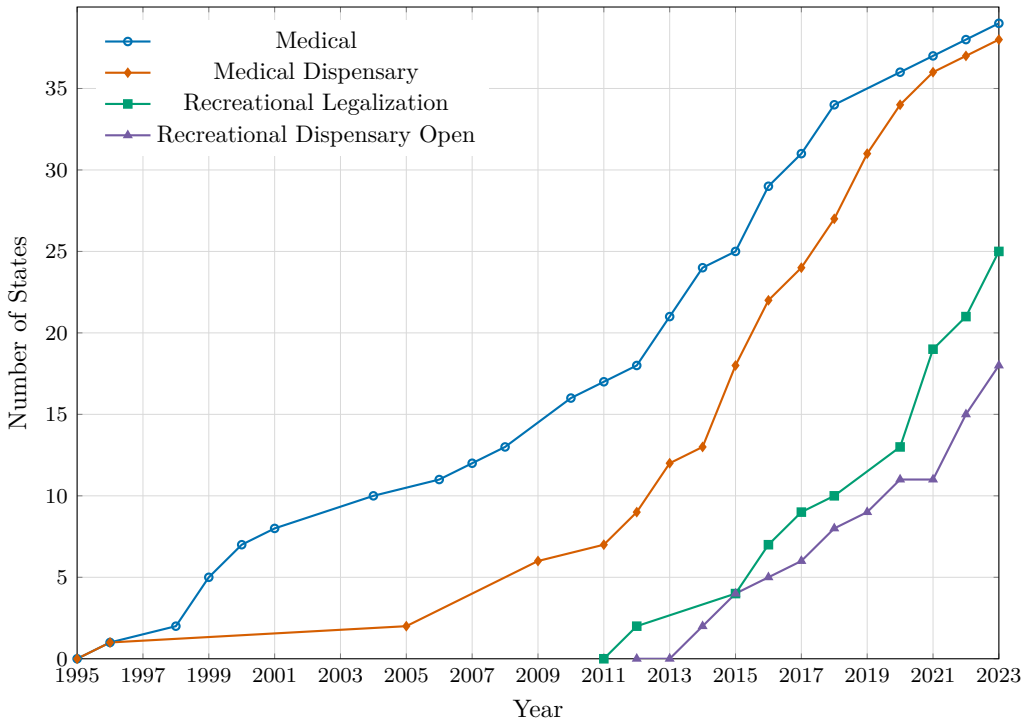
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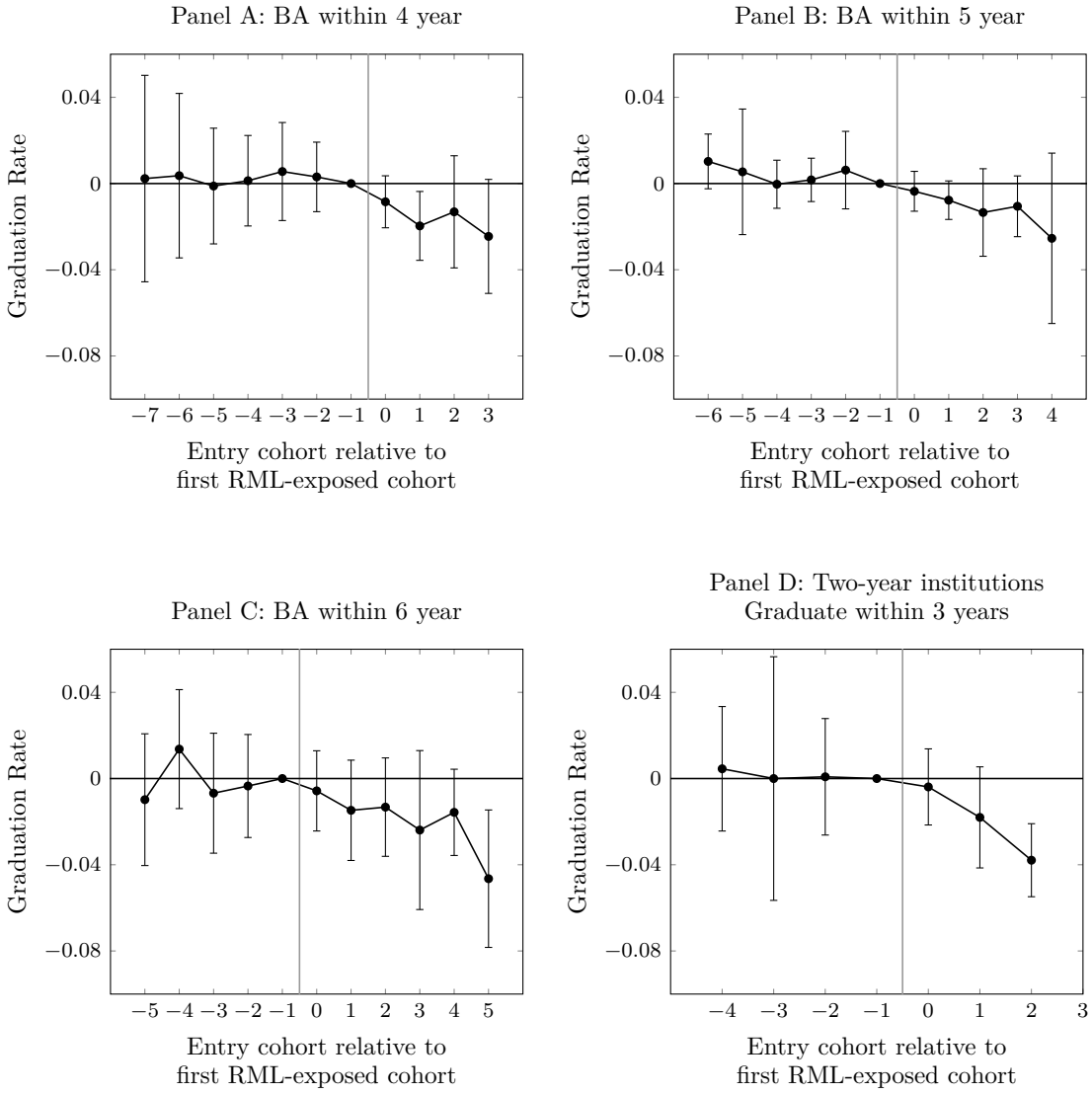
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Figure 1: Number of States with Marijuana Policy Over Time



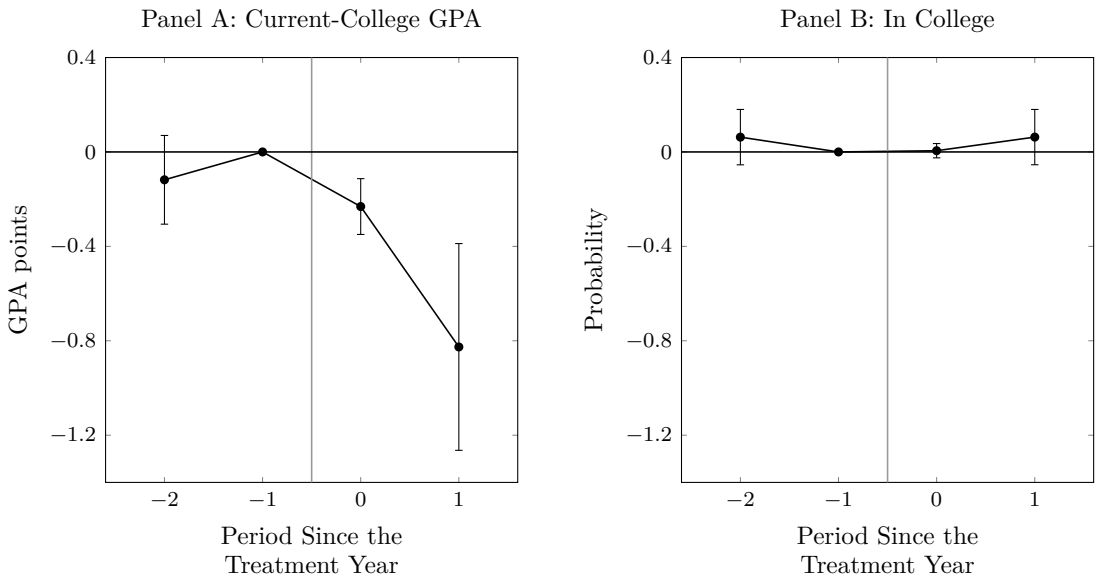
Note: 1995 to 2023 counts are based on policy adoption records from state legislative sources.

Figure 2: RML Event Study Plots for Graduation Rate



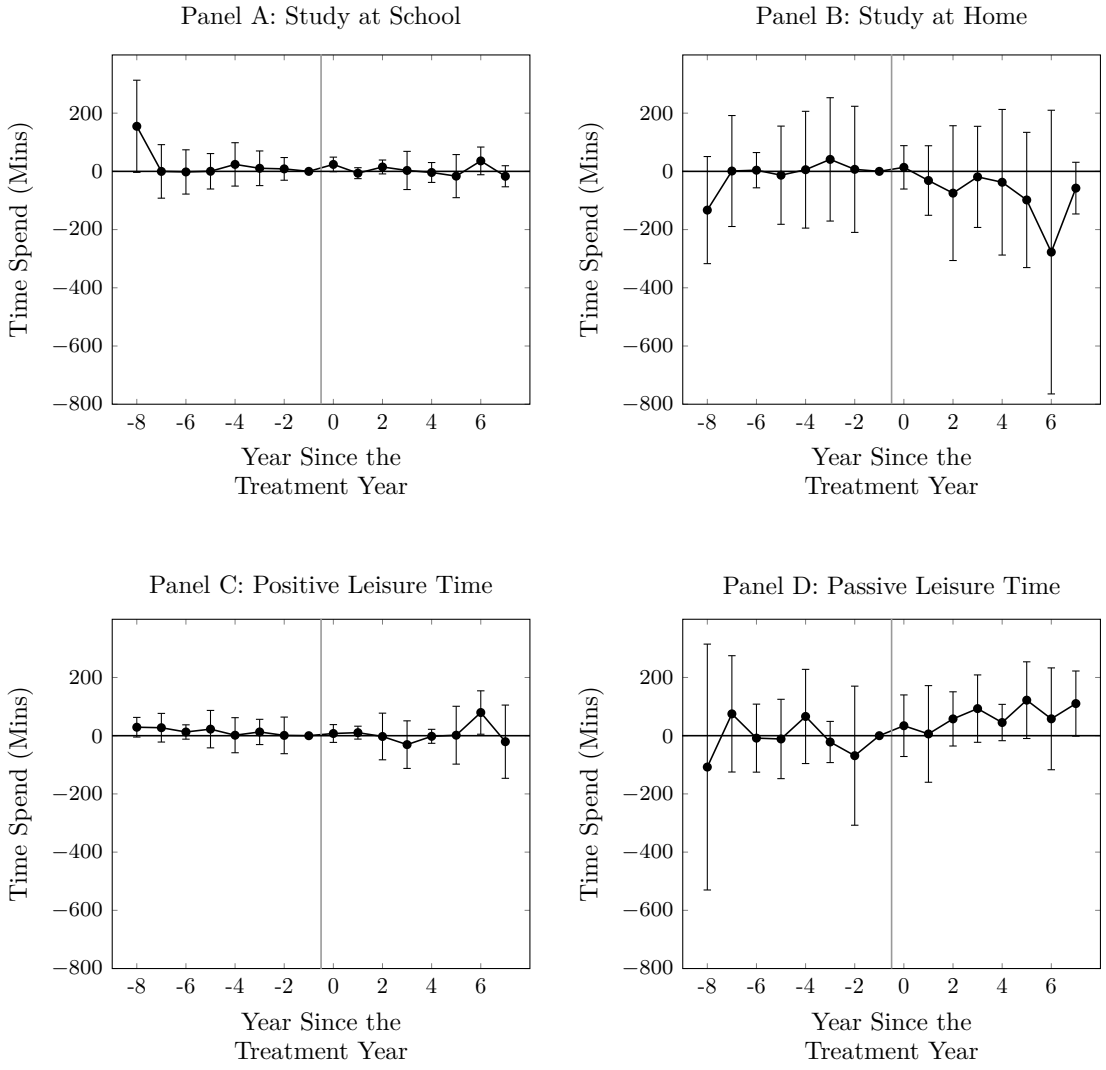
Note: Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level. Panel A,B and C are based on the 4 year institutions, while panel D is based on the 2 year institution. The graduation rate is calculated based on full-time, first-time degree-seeking undergraduate students. Not-yet treated units are used as the control group. Entry-cohort event time is zero for the first cohort whose completion window reaches RML; −1 is the reference cohort. The supported event-time range differs across completion horizons.

Figure 3: RML Event-Study Estimates for GPA and College Attendance



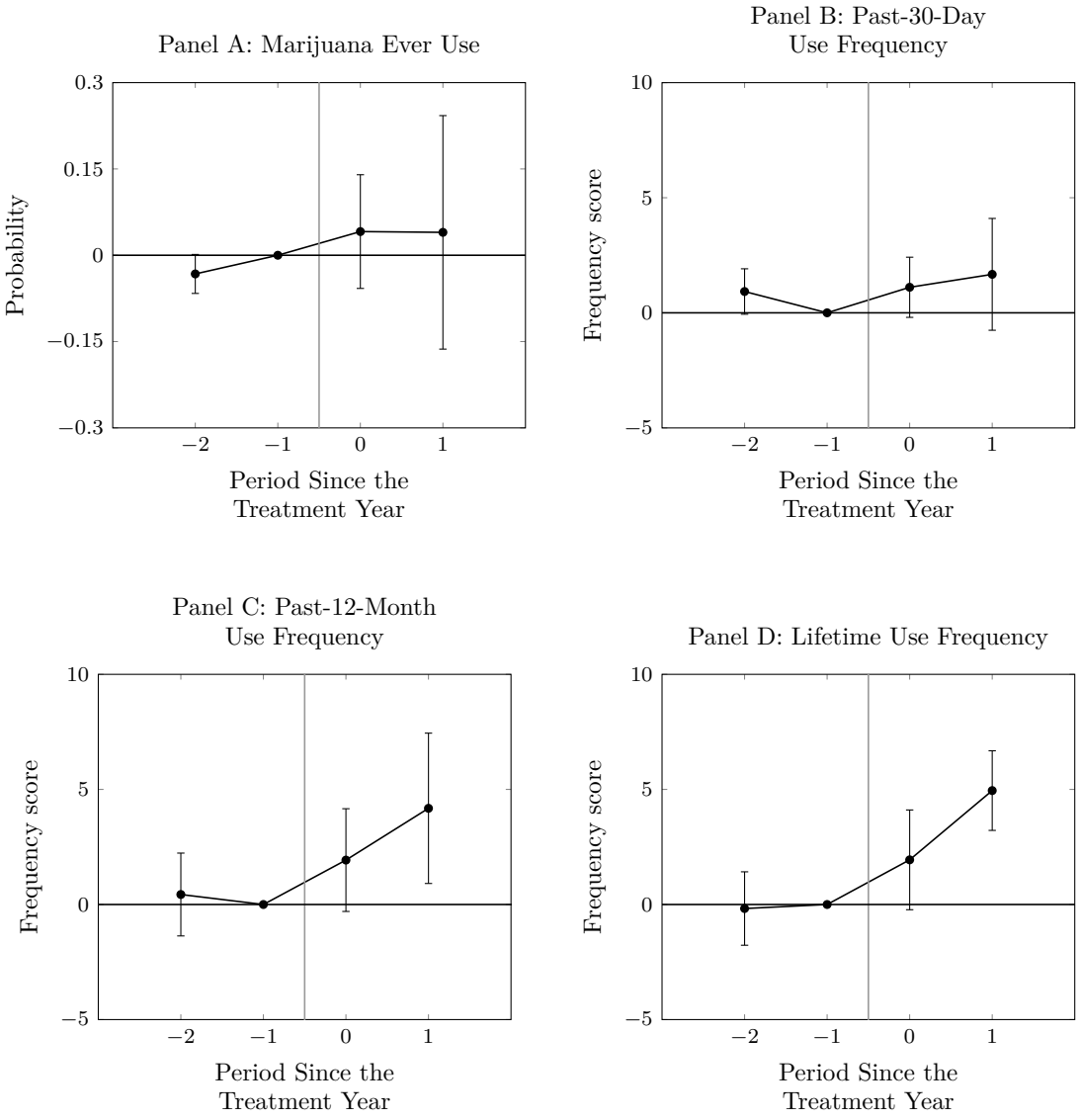
Note: Both samples are restricted to ages 18–24. Panel A uses GPA among currently enrolled students; Panel B uses the available 2011–2015 enrollment series. Each period is two years, and period −1 is the reference. Individual and period fixed effects are included; not-yet-treated units are controls. Points are Callaway and Sant'Anna (2021) estimates, with 95% intervals from state-clustered standard errors. Both panels show periods −2 through 1 for comparison. The earlier enrollment lead at −3, outside this common display window, is negative and statistically significant; the enrollment evidence should therefore be interpreted cautiously.

Figure 4: RML Event Study Plots for Time Use



Note: The not-yet treated groups are used as the control group. Individual age is from 18 to 24. Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level.

Figure 5: RML Event Study Estimates for Marijuana Use



Note: Panel A present the event-study estimates for marijuana ever use, while panels B–D present the corresponding event-study estimates for marijuana use frequency. All specifications include individual and period fixed effects. Each period represents two years. Not-yet-treated groups are used as the control group. The sample is restricted to individuals ages 18–24. Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level.

Table 1: Summary Statistics, Before and After RML

	All	Pre treated	Post treated	Observations
	(1)	(2)	(3)	(4)
NCES				
Graduate with BA in 4 Year	0.430	0.419	0.452	2752
	(0.224)	(0.226)	(0.218)	
Graduate with BA in 5 Year	0.561	0.543	0.576	2707
	(0.194)	(0.197)	(0.190)	
Graduate with BA in 6 Year	0.599	0.584	0.611	2707
	(0.181)	(0.182)	(0.179)	
Graduate from 2 Year Institution in 3 Year	0.354	0.336	0.378	1841
	(0.207)	(0.209)	(0.203)	
PSID				
GPA	3.24	3.25	3.21	568
	(0.493)	(0.482)	(0.505)	
In College	0.438	0.446	0.385	788
	(0.496)	(0.497)	(0.489)	
Marijuana ever	0.534	0.498	0.577	1546
	(0.499)	(0.500)	(0.494)	
Marijuana Freq. (30d)	4.929	4.049	5.987	1540
	(11.499)	(10.561)	(12.461)	
Marijuana Freq. (12m)	8.367	7.058	9.934	1544
	(15.042)	(14.054)	(16.015)	
Marijuana Freq. (Life)	12.964	11.493	14.729	1542
	(17.160)	(16.618)	(17.639)	
ATUS				
Study Time at School (min)	28.28	29.20	26.79	2042
	(91.11)	(92.37)	(89.04)	
Study Time at Home (min)	42.74	37.39	51.45	2042
	(109.80)	(97.83)	(126.47)	
Positive Leisure time (min)	33.90	32.14	36.77	2042
	(83.98)	(81.79)	(87.41)	
Passive Leisure time (min)	194.18	191.46	198.61	2042
	(180.90)	(176.43)	(187.99)	

Note: This table reports summary statistics for all main variables in the RML analytic sample, consisting of the 15 states that had enacted MML by 2010. In the NCES data, BA graduation outcomes are constructed from cohorts entering between 2006 and 2017, and three-year graduation outcomes at 2-year institutions are constructed from cohorts entering between 2008 and 2017. PSID measures use the 2011–2023 waves, except In College, which is available only in 2011, 2013, and 2015. GPA is observed among currently enrolled students; In College includes both enrollment statuses. In the ATUS, all variables are based on the 2010–2023 waves. Individuals in both the PSID and ATUS samples are restricted to ages 18–24.

Frequency statistics use valid midpoint-coded scores; observations with unknown use are excluded. Descriptive counts refer to available observations, rather than the subset used in the corresponding regression.

Table 2: States with MML Treatment Year Before 2010, by 2023

State	RML Treatment Year	MML Treatment Year
Colorado	2013	2001
Washington	2013	1999
Alaska	2015	1999
Washington, DC	2015	2010
Oregon	2015	1999
Maine	2017	2000
California	2017	1997
Nevada	2017	2002
Vermont	2018	2004
Michigan	2019	2009
Montana	2021	2005
New Jersey	2021	2010
New Mexico	2021	2007
Rhode Island	2022	2006
Hawaii	.	2000

Note: State treatment years are defined such that enactments occurring on or after September 1 are assigned to the subsequent calendar year.

Table 3: Estimates of RML on Education Attainment

	4 Year Institution			2 Year Institution
	Within 4 year	Within 5 year	Within 6 year	Within 3 year
	(1)	(2)	(3)	(4)
ATT	-0.016^{***}	-0.011^{**}	-0.018^{***}	-0.020^{***}
	(0.006)	(0.005)	(0.005)	(0.006)
Pre Means	0.419	0.548	0.584	0.336
Obs.	2,752	2,707	2,707	1,841

Note: Columns 1–3 present the estimated effects of RML on BA completion within four, five, and six years among students at 4-year institutions. Column 4 presents the estimated effect of RML on the three-year graduation rate at 2-year institutions. Institution and time fixed effects are included. The not-yet treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Obs. reports institution–entry-cohort observations used in each estimation.

Table 4: Estimates of RML Effects on GPA and College Attendance

	GPA (1)	In College (2)
ATT	-0.350^{***}	0.020
	(0.095)	(0.021)
Pre Means	3.33	0.474
Obs.	217	627

Note: Both samples are restricted to ages 18–24. GPA is measured among currently enrolled college students. In College includes enrolled and non-enrolled respondents. The GPA source covers the 2011–2023 waves; observations used in estimation span 2011–2019 for GPA and 2011–2015 for In College. Individual and time fixed effects are included; not-yet-treated units are controls. Obs. reports the number of person-waves used in estimation. Pre means use untreated observations, including never-treated units, within each estimation sample. Standard errors in parentheses and significance stars use state-clustered wild bootstrap inference. *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

Table 5: Estimates of RML Effects on Time Use

	Study at School	Study at Home	Positive Leisure Time	Passive Leisure Time
	(1)	(2)	(3)	(4)
ATT	5.53	−44.43***	0.750	54.17**
	(7.84)	(8.53)	(9.30)	(23.88)
Pre Means	29.20	37.39	32.14	191.46
Obs.	1,924	1,924	1,924	1,924

Note: State and time fixed effects are included. The not-yet treated groups are used as the control group. Individual age is from 18 to 24. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.
Obs. reports the number of respondent-day observations used in each estimation.

Table 6: Estimates of RML Effects on Marijuana Use

	Ever	Frequency		
		In 30 days	In 12 months	In Lifetime
	(1)	(2)	(3)	(4)
ATT	0.041	1.26**	2.54***	2.75***
	(0.027)	(0.614)	(0.667)	(0.857)
Pre Means	0.468	4.37	6.97	10.59
Obs.	867	866	866	866

Note: Individual and time fixed effects are included. The not-yet treated groups are used as the control group. Individual age is from 18 to 24. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Appendix A: Supplementary Data and Robustness Details

This appendix describes survey design and variable coding, estimation and policy-timing details, time-use classification, reporting-bias checks, sample construction under alternative MML cutoffs, and the effects of MML on BA completion.

A1 Survey Design and Measurement

IPEDS is effectively a census of postsecondary institutions that participate in federal student aid programs and provides annual institution-level information on enrollment, student composition, admissions, academic offerings, and degree completion. For a given cohort, the four-year, five-year, and six-year bachelor's degree completion rates are measured in the fourth, fifth, and sixth years after initial enrollment, respectively, while the three-year graduation rate for two-year institutions is measured in the third year after initial enrollment. IPEDS-defined exclusions include students who left the institution because they died or became permanently disabled, entered the armed forces, joined a federal foreign aid service such as the Peace Corps, or participated in official church missions.

I use the 2021 Carnegie Basic Classification throughout. R1 and R2 institutions are grouped as *Doctoral*; Master's institutions with larger, medium, and small programs are grouped as *Master*; all Associate's, Baccalaureate/Associate's, and Baccalaureate institutions are grouped as *Bacc/Assoc*; and all remaining institutions are classified as *Other*. This classification is not available for two-year institutions.

To examine intermediate effects on students' time allocation, I use the ATUS data, which is an annual, nationally representative repeated cross-sectional survey conducted by the U.S. Bureau of Labor Statistics in collaboration with the U.S. Census Bureau. In each survey year, one randomly selected adult from each sampled household is interviewed and asked to complete a detailed 24-hour diary describing activities undertaken on the previous day, including the timing, location, and social context of each activity.

The PSID is a nationally representative longitudinal survey that has followed U.S. individuals and families since 1968, collecting rich information on demographics, socioeconomic status, health, and behavioral outcomes. The survey re-interviews the same individuals biennially, making it possible to track outcomes over time.

Use frequency is based on respondents' reported number of occasions of use in the past 30 days, past 12 months, and over the lifetime. Because the response categories are reported in intervals (1–2, 3–5, 6–9, 10–19, 20–39, and 40 or

more times), I recode them to midpoint values. For example, I code 6–9 times as 7.5 and 40 or more times as 40. These values are frequency scores rather than exact occasion counts. A composite frequency score sums the scores for its component substances and is observed only when every component is valid. Unknown responses and unavailable items remain missing; only valid nonuse responses and their documented skips are coded zero.

Drink quantity instead uses the reported number of drinks on drinking days, with one drink or fewer coded as one. I include zeros only when the response path establishes no drinking, and retain unknown or inconsistent responses as missing. The measure includes verified non-drinking zeros and is not average consumption per calendar day.[24]

A2 Estimation and Policy-Timing Details

The Callaway and Sant'Anna (2021) framework readily accommodates event-study analyses, unbalanced panels, and repeated cross-sectional data, all of which are relevant to my setting. An additional practical advantage is that the framework provides a built-in wild bootstrap procedure for inference. Implemented as a one-step wild (multiplier) bootstrap, this procedure is computationally efficient relative to a conventional resampling bootstrap that re-estimates the full model in every replication.

The lack of statistical significance for individual event-time coefficients does not necessarily imply the absence of treatment effects. Each coefficient is estimated with less precision because it relies on a narrower subset of the data and captures a period-specific dynamic effect. By contrast, the overall ATT aggregates information across all post-treatment periods, thereby increasing statistical power and yielding a more precisely estimated average treatment effect. This feature becomes even more pronounced under wild bootstrap inference, which produces more conservative confidence intervals for individual dynamic coefficients, further reducing the likelihood that any single event-time estimate is statistically significant even when the average post-treatment effect is statistically distinguishable from zero.

For bachelor's degree completion outcomes, I restrict attention to cohorts exposed to RML between academic years 1 and 11 after first enrollment. The omitted reference cohort is academic year 5 for four-year completion, academic year 6 for five-year completion, and academic year 7 for six-year completion. For the three-year graduation rate at two-year institutions, I restrict attention

[24]In 2023, the extracted data do not include the additional lifetime-drinking gate needed to distinguish some non-drinking skips from unknown responses. These ambiguous observations remain missing.

to cohorts exposed between academic years 1 and 7 and use academic year 4 as the omitted category.

I recode state treatment years so that enactments occurring on or after September 1 are assigned to the subsequent calendar year. For example, Washington and Colorado—the first two states to legalize RML—enacted their laws in late 2012; accordingly, I code their treatment year as 2013. Therefore, the composition of all students who entered before 2013 is unaffected by RML.

To assess the identifying assumption that the timing of legalization is unrelated to the main outcomes, Table A1 reports regressions in which the legalization timing of the sample states is used as the dependent variable and each main outcome is entered separately as an explanatory variable. Robust standard errors are reported, and 95% confidence intervals are constructed using the wild bootstrap with 999 repetitions. For BA graduation outcomes, I use the 2006 entering cohort; for graduation outcomes at two-year institutions, I use the 2008 entering cohort; for GPA, substance use and other health behavior, I use the 2011 sample; and for time use, I use the 2010 sample.

A3 Time-Use Activity Classification

Table A11 lists the ATUS codes used to construct the time-use measures. Following the activity classification in Chu and Gershenson (2018), education-related time consists of attending class, participating in non-sport extracurricular school activities, and engaging in research or homework. Leisure time is measured using the "relaxing and leisure" category. Positive leisure time includes relaxing and thinking, arts and other personal-interest activities, other relaxing and leisure activities, and out-of-home leisure such as visiting museums and going to the movies. Passive leisure time includes activities such as smoking, drug use, watching television, listening to the radio, and playing computer games.

A4 Reporting-Bias Checks

When measuring ever use of substances, drinking, and smoking, two potential sources of reporting bias should be considered. The first arises because, relative to MML, RML may affect higher educational outcomes partly by changing social norms and perceptions of risk. A natural concern, therefore, is that individuals may become more willing to disclose marijuana or other illicit drug use after RML. A second source of concern is inconsistency in self-reports over time.

Table A12 reports the placebo tests for changes in disclosure after RML. I define reporting bias as occurring when an individual does not report ever using a substance, drinking, or smoking before legalization, but does report such

use after legalization, even though the reported age at first use is lower than the individual's age at the time legalization occurred in his or her state. To assess whether this reporting-bias measure is statistically meaningful, I conduct placebo tests. For each substance-use outcome, I run 1,000 placebo simulations in which RML legalization years are randomly reassigned across states.

Table A17 reports the checks for inconsistent self-reports discussed in Section 7. This issue arises when an individual reports ever use of a substance, drinking, or smoking in one interview, but later denies such use in at least one subsequent interview. Unlike the first type of reporting bias, this second type is unlikely to be driven by legalization itself; instead, it more plausibly reflects measurement error or respondent recall problems. In Column (1), for the ever-use measures, I impose consistency for individuals who initially reported use but later reported no use, such that once an individual first reports use, all subsequent ever-use observations are coded as use. In Column (2), these inconsistent individuals are instead excluded from the construction of the ever-use measures. Columns (3)-(6) report the frequency and quantity results using the sample that excludes these inconsistent individuals.

A5 Alternative MML Cutoffs

The main analysis includes 15 states (Table 2). Together, they account for 26.58% of the total U.S. population in 2023, based on the U.S. Census Bureau's July 1, 2023 state population estimates, and 50.08% of the population living in states that had legalized recreational marijuana by 2023.

Tables A4–A10 report the estimates under alternative MML cutoff years discussed in Section 7. This subsection provides the corresponding sample-construction details.

Table A4 reports graduation-rate estimates under alternative MML cutoff years. For each specification, the cohort enrollment range is defined as (cutoff year − 4, 2017) when examining BA attainment, and as (cutoff year − 3, 2017) when examining graduation rates at 2-year institutions, in order to ensure that the pre-treatment period for all included cohorts is not affected by MML.[25]

Tables A5–A10 report the corresponding estimates for the other outcomes. Unlike the graduation-rate analysis, varying the cutoff year in these specifications

[25]Because the sample for BA graduation outcomes includes cohorts entering college between 1996 and 2017, the estimation of RML effects on BA attainment can incorporate as many cohorts as possible. By contrast, the sample for graduation rates at 2-year institutions includes only cohorts entering between 2008 and 2017. Therefore, when the cutoff year is earlier than 2011, all regressions for 2-year institution outcomes are estimated using the same cohort range, namely 2008–2017.

changes only the composition of states included in the sample and does not alter the underlying data period. Specifically, the time-use analysis continues to use data from 2010 to 2023, and the GPA and substance-use analyses continue to use the 2011–2023 waves of the PSID (2011, 2013, ..., 2023).

A6 MML and BA Completion

I also estimate the effects of MML on BA completion using IPEDS data from 1997 to 2012. I apply the same sample-screening procedures and cohort-based difference-in-differences approach as in the RML analysis, with treatment timing defined by MML adoption. Table A18 reports the results for four-year institutions. Insufficient data for the earlier years prevent a corresponding analysis of graduation rates at two-year institutions.

MML reduces four-year BA completion by 3.15 percentage points and increases five-year completion by 2.84 percentage points, with the estimates significant at the 1% and 5% levels, respectively. The six-year estimate is positive but statistically insignificant. This pattern is consistent with delayed degree completion, although the imprecise six-year estimate does not establish that longer-run attainment is unaffected.

Table A1: Timing of RML Adoption and Main Results

Variable	Coefficient	Std. Error	t	95% CI by Wild Bootstrap	Obs.	Clusters
NCES						
Graduate with BA in 4 year	1.29	1.16	1.12	[-1.05, 5.91]	189	9
Graduate with BA in 5 year	0.494	0.819	0.603	[-2.98, 2.76]	180	8
Graduate with BA in 6 year	0.746	0.957	0.779	[-3.14, 3.22]	180	8
Graduate from 2 year institution in 3 year	-0.970	0.381	-2.55	[-1.76, 2.85]	49	4
PSID						
GPA	-0.086	0.406	-0.212	[-1.24, 3.70]	79	6
Marijuana ever use	-0.254	0.349	-0.728	[-1.67, 1.16]	237	6
Marijuana frequency past 30 days	-0.00648	0.0087	-0.745	[-0.0546, 0.0224]	237	6
Marijuana frequency past 12 months	-0.0222	0.014	-1.59	[-0.0548, 6.75e-05]	237	6
Marijuana frequency lifetime	-0.0117	0.0127	-0.922	[-0.0417, 0.0492]	237	6
Ever Drink	-0.065	0.398	-0.163	[-1.45, 2.21]	356	10
Drinks per drinking day	-0.00874	0.0752	-0.116	[-0.234, 0.453]	352	10
Ever Smoke	-0.094	0.394	-0.239	[-1.79, 0.803]	356	10
Cocaine ever use	-0.479	0.498	-0.963	[-1.54, 1.43]	237	6
Cocaine frequency past 30 days	-0.125	0.217	-0.576	[-6.59, 123]	237	6
Cocaine frequency past 12 months	-0.0216	0.0322	-0.669	[-0.342, 6.07]	237	6
Cocaine frequency lifetime	-0.0322	0.0238	-1.35	[-0.0661, 0.0993]	237	6

Table A1: Timing of RML Adoption and Main Results (continued)

Variable	Coefficient	Std. Error	t	95% CI by Wild Bootstrap	Obs.	Clusters
Group A ever use	-0.754	0.197	-3.83	[-1.02, -0.065]	237	6
Group A frequency past 30 days	-0.0271	0.0261	-1.04	[-0.287, 2.31]	237	6
Group A frequency past 12 months	-0.00282	0.00579	-0.488	[-0.0293, 0.026]	236	6
Group A frequency lifetime	-0.0117	0.00246	-4.77	[-0.0181, 0.0073]	237	6
Group B ever use	0.354	0.357	0.992	[-0.303, 2.13]	237	6
Group B frequency past 30 days	0.0504	0.0416	1.21	[-33.5, 18.7]	237	6
Group B frequency past 12 months	0.0581	0.0513	1.13	[-11.7, 7.71]	237	6
Group B frequency lifetime	-0.0036	0.00626	-0.575	[-0.0173, 0.0402]	237	6
ATUS						
Study Time at School (min)	-0.000	0.001	-0.192	[-0.004, 0.016]	213	8
Study Time at Home (min)	0.001	0.001	1.01	[-0.001, 0.013]	213	8
Positive Leisure time (min)	-0.001	0.001	-0.441	[-0.012, 0.004]	213	8
Passive Leisure time (min)	-0.000	0.000	-1.02	[-0.002, 0.001]	213	8

Note: This table reports the coefficient from a regression of RML treatment year on the main variables. The bracketed values are the 95% wild bootstrap confidence interval obtained using wild bootstrap with clustering at the state level. Group A comprises amphetamines, barbiturates, and tranquilizers; Group B comprises diet pills and steroids. Obs. reports the number of observations used in each policy-timing regression.

Table A2: Effective Dates of Marijuana Laws, by 2023

State	**Medical**	**Medical Dispensary**	**Recreational**	**Recreational Dispensary**
Alabama	May 17, 2021	Sept 1, 2022	.	.
Alaska	Mar 2, 1999	Oct 1, 2016	Feb 23, 2015	Oct 29, 2016
Arizona	Dec 14, 2010	Dec 6, 2012	Nov 30, 2020	Jan 22, 2021
Arkansas	Nov 9, 2016	May 11,2019	.	.
California	Nov 6, 1996	Nov 10, 1996	Nov 9, 2016	Jan 1, 2018
Colorado	Dec 28, 2000	Oct 1, 2005	Dec 10, 2012	Jan 1, 2014
Connecticut	Oct 1, 2012	Aug 20, 2014	June 22, 2021	Jan 10, 2023
Delaware	Jul 1, 2011	Jun 26, 2015	Apr 23, 2023	.
Washington, DC	Jul 27, 2010	Jul 29, 2013	Feb 26, 2015	.
Florida	Nov 8, 2016	Jul 19, 2016	.	.
Georgia	.	.	.	.
Hawaii	Jun 14, 2000	Aug 8, 2017	.	.
Idaho	.	.	.	.
Illinois	Aug 1, 2013	Aug 25, 2015	Jan 1, 2020	Jan 1, 2020
Indiana	.	.	.	.
Iowa	.	.	.	.
Kansas	.	.	.	.
Kentucky	Mar 31, 2023	.	.	.
Louisiana	June 30, 2015	Aug 6, 2019	.	.
Maine	Nov 2, 1999	Mar 31, 2011	Nov 8, 2016	Oct 9, 2020
Maryland	Apr 14, 2014	Jul 5, 2017	Jul 1, 2023	Jul 1, 2023
Massachusetts	Jan 1, 2013	Jun 24, 2015	Dec 15, 2016	Nov 20, 2018
Michigan	Nov 4, 2008	July 15, 2018	Dec 6, 2018	Dec 1, 2019
Minnesota	May 29, 2014	Jul 1, 2015	Aug 1, 2023	.
Mississippi	Feb 2, 2022	Jan 25, 2023	.	.
Missouri	Dec 6, 2018	Oct 17, 2020	Dec 8, 2022	Feb 3, 2023

Table A2: Effective Dates of Marijuana Laws, by 2023 (continued)

State	Medical	Medical Dispensary	Recreational	Recreational Dispensary
Montana	Nov 2, 2004	Jan 1, 2009	Jan 1, 2021	Jan 1, 2022
Nebraska	.	.	.	.
Nevada	Oct 1, 2001	Jul 31, 2015	Jan 1, 2017	July 1, 2017
New Hampshire	Jul 23, 2013	Apr 30, 2016	.	.
New Jersey	Jan 18, 2010	Dec 6, 2012	Jan 1, 2021	Apr 21, 2022
New Mexico	Jul 1, 2007	Jul 1, 2009	June 29, 2021	Apr 1, 2022
New York	Jul 5, 2014	Jan 7, 2016	Mar 30, 2021	Dec 29, 2022
North Carolina	.	.	.	.
North Dakota	Nov 8, 2016	Feb 25, 2019	.	.
Ohio	June 8, 2016	Jan 16, 2019	Dec 7, 2023	.
Oklahoma	June 26, 2018	June 27, 2018	.	.
Oregon	Mar 4, 1999	Nov 14, 2009	Jul 1, 2015	Oct 1, 2015
Pennsylvania	May 17, 2016	Jan 17, 2018	.	.
Rhode Island	Jan 3, 2006	Apr 19, 2013	May 25, 2022	.
South Carolina	.	.	.	.
South Dakota	Nov 3, 2020	Jul 27, 2021	.	.
Tennessee	.	.	.	.
Texas	.	.	.	.
Utah	Dec 1, 2018	Mar 2, 2020	.	.
Vermont	Jul 1, 2004	Jun 21, 2013	Jul 1, 2018	.
Virginia	Jul 1, 2020	Oct 17, 2020	Jul 1, 2021	.
Washington	Dec 3, 1998	Oct 1, 2009	Dec 6, 2012	Jul 8, 2014
West Virginia	Apr 19, 2017	Nov 2, 2021	.	.
Wisconsin	.	.	.	.
Wyoming	.	.	.	.

Note: Dates are from Guernsey, Serfling and Yan (2024), Cheng, De Franco and Lin (2023) ,https://www.procon.org, and internet searches

Table A3: Summary Statistics for Other Health Behavior, Before and After RML

	All	Pre treated	Post treated	Observations
	(1)	(2)	(3)	(4)
PSID				
Ever Drink Alcohol	0.616	0.646	0.594	2361
	(0.486)	(0.478)	(0.491)	
Drinks per drinking day	2.393	2.177	2.574	2180
	(3.311)	(2.536)	(3.830)	
Ever Smoke Cigarettes	0.223	0.236	0.204	1710
	(0.416)	(0.425)	(0.403)	
Cocaine Ever	0.102	0.089	0.116	1546
	(0.302)	(0.285)	(0.321)	
Cocaine Freq. (30d)	0.054	0.041	0.070	1546
	(0.415)	(0.386)	(0.446)	
Cocaine Freq. (12m)	0.307	0.255	0.369	1546
	(1.939)	(1.920)	(1.962)	
Cocaine Freq. (Life)	1.080	0.863	1.340	1546
	(5.095)	(4.528)	(5.692)	
Group A Ever	0.167	0.178	0.153	1546
	(0.373)	(0.383)	(0.361)	
Group A Freq. (30d)	0.729	0.802	0.642	1546
	(4.466)	(4.772)	(4.071)	
Group A Freq. (12m)	1.723	2.048	1.335	1545
	(8.382)	(9.635)	(6.566)	
Group A Freq. (Life)	4.245	4.817	3.559	1545
	(13.981)	(15.230)	(12.294)	
Group B Ever	0.109	0.127	0.087	1546
	(0.311)	(0.333)	(0.282)	
Group B Freq. (30d)	0.205	0.267	0.131	1546
	(2.320)	(2.585)	(1.957)	
Group B Freq. (12m)	0.483	0.650	0.283	1545
	(3.667)	(4.185)	(2.920)	
Group B Freq. (Life)	1.304	1.493	1.076	1545
	(6.098)	(6.533)	(5.527)	

Note: PSID respondents aged 18–24 in the 15 states with MML by 2010; waves 2011–2023. Counts are available valid observations. Group A comprises amphetamines, barbiturates, and tranquilizers; Group B comprises diet pills and steroids. Some items include prescribed use. Composite frequency scores sum valid component scores and require every component to be observed. Drink quantity records drinks on drinking days, with verified non-drinking zeros; one drink or fewer is coded as one. Unknown or inconsistent responses are missing. See Appendix A1.

Table A4: Robustness Results of Graduation Rate

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Panel A. 4-Year Institutions							
Within 4	-0.023^{***}	-0.021^{***}	-0.021^{***}	-0.021^{***}	-0.026^{***}	-0.009	-0.005
	(0.002)	(0.003)	(0.003)	(0.003)	(0.005)	(0.009)	(0.008)
	[0.419]	[0.410]	[0.409]	[0.409]	[0.418]	[0.414]	[0.430]
Obs.	1,472	1,681	1,725	1,758	1,851	1,989	2,166
Within 5	-0.030^{***}	-0.030^{***}	-0.031^{***}	-0.032^{***}	-0.031^{***}	-0.014	-0.009
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.008)	(0.009)
	[0.551]	[0.542]	[0.542]	[0.542]	[0.548]	[0.544]	[0.548]
Obs.	1,472	1,681	1,725	1,758	1,851	1,989	2,085
Within 6	-0.025^{***}	-0.023^{***}	-0.024^{***}	-0.024^{***}	-0.022^{***}	-0.022^{***}	-0.022^{***}
	(0.002)	(0.005)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
	[0.594]	[0.577]	[0.577]	[0.577]	[0.584]	[0.580]	[0.585]
Obs.	1,598	1,871	1,881	1,878	1,931	2,031	2,085
Institutions	132	151	155	158	166	181	193
Panel B. 2-Year Institutions							
Within 3	-0.025^{***}	-0.026^{***}	-0.027^{***}	-0.027^{***}	-0.030^{***}	-0.029^{***}	-0.029^{***}
	(0.003)	(0.003)	(0.003)	(0.003)	(0.008)	(0.007)	(0.007)
	[0.350]	[0.356]	[0.359]	[0.359]	[0.358]	[0.354]	[0.353]
Obs.	1,188	1,258	1,272	1,272	1,285	1,359	1,365
Institutions	170	184	186	186	187	195	196
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Institution and time fixed effects are included. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Obs. reports institution–entry-cohort observations used in the corresponding estimation.

Table A4: Robustness Results of Graduation Rate (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Panel A. 4-Year Institutions						
Within 4	−0.011	−0.011	−0.012	−0.016***	−0.018***	−0.017***
	(0.009)	(0.009)	(0.008)	(0.006)	(0.006)	(0.005)
	[0.424]	[0.425]	[0.412]	[0.419]	[0.415]	[0.410]
Obs.	2,181	2,137	2,475	2,752	2,713	2,540
Within 5	−0.011	−0.011	−0.010	−0.011**	−0.012*	−0.012*
	(0.009)	(0.009)	(0.006)	(0.005)	(0.006)	(0.007)
	[0.543]	[0.544]	[0.540]	[0.548]	[0.542]	[0.537]
Obs.	2,109	2,074	2,421	2,707	2,677	2,313
Within 6	−0.022***	−0.024***	−0.018*	−0.018***	−0.019**	−0.018**
	(0.005)	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)
	[0.580]	[0.581]	[0.578]	[0.584]	[0.578]	[0.572]
Obs.	2,109	2,074	2,421	2,707	2,417	2,297
Institutions	201	202	243	285	302	306
Panel B. 2-Year Institutions						
Within 3	−0.029***	−0.029***	−0.020***	−0.020***	−0.023***	−0.024***
	(0.006)	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)
	[0.347]	[0.347]	[0.333]	[0.336]	[0.335]	[0.338]
Obs.	1,512	1,512	1,673	1,841	1,979	1,964
Institutions	217	217	240	264	284	290
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Institution and time fixed effects are included. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Obs. reports institution–entry-cohort observations used in the corresponding estimation.

Table A5: Robustness Results of GPA and Time Use

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Panel A. GPA							
GPA	0.013	−0.230	−0.182	−0.182	−0.182	−0.182	−0.182
	(0.012)	(0.173)	(0.164)	(0.164)	(0.164)	(0.164)	(0.157)
	[3.19]	[3.22]	[3.24]	[3.24]	[3.24]	[3.24]	[3.24]
Obs.	87	102	107	107	107	107	107
Panel B. Time Use							
Study at	3.03	1.71	0.479	0.479	0.389	18.07	−4.37
School	(1.80)	(1.07)	(6.68)	(6.68)	(4.19)	(14.41)	(8.30)
	[23.94]	[24.07]	[25.88]	[25.88]	[25.79]	[26.71]	[26.88]
Obs.	1,066	1,155	1,222	1,222	1,225	1,400	1,411
Study at	−18.09	−34.82*	−35.00***	−35.00***	−35.00***	−104.00	−93.10
Home	(10.56)	(10.46)	(6.20)	(6.20)	(6.28)	(51.05)	(66.30)
	[35.48]	[37.13]	[36.63]	[36.63]	[36.55]	[37.48]	[37.71]
Obs.	1,066	1,155	1,222	1,222	1,225	1,400	1,411
Positive	4.74	3.63	3.81	3.81	3.84	6.96	11.50***
Leisure	(7.67)	(5.34)	(9.03)	(9.03)	(6.64)	(4.41)	(4.46)
	[31.84]	[32.31]	[32.07]	[32.07]	[31.95]	[32.05]	[31.72]
Obs.	1,066	1,155	1,222	1,222	1,225	1,400	1,411
Passive	83.63***	−3.25	−0.199	−0.199	−0.878	27.68	43.50
Leisure	(19.57)	(42.81)	(46.49)	(46.49)	(48.11)	(28.34)	(25.76)
	[193.67]	[193.53]	[191.26]	[191.26]	[191.67]	[191.68]	[191.61]
Obs.	1,066	1,155	1,222	1,222	1,225	1,400	1,411
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. Panel A measures GPA among currently enrolled students and includes individual and time fixed effects. Its pre means and Obs. refer to the corresponding estimation sample; Obs. counts person-waves. Individual age is from 18 to 24. Panel B includes state and time fixed effects. The pre-treatment mean is reported in square brackets. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

In Panel B, Obs. reports respondent-day observations used in the corresponding time-use estimation.

Table A5: Robustness Results of GPA and Time Use (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Panel A. GPA						
GPA	−0.182	−0.182	−0.143**	−0.350***	−0.352***	−0.352***
	(0.157)	(0.157)	(0.080)	(0.095)	(0.089)	(0.089)
	[3.24]	[3.24]	[3.30]	[3.33]	[3.32]	[3.32]
Obs.	107	107	171	217	217	217
Panel B. Time Use						
Study at	7.51	7.51	17.15*	5.53	−2.68	−2.68
School	(78.91)	(78.91)	(8.15)	(7.84)	(8.92)	(8.92)
	[26.47]	[26.47]	[29.24]	[29.20]	[29.55]	[29.55]
Obs.	1,458	1,458	1,698	1,924	2,121	2,121
Study at	−18.90	−18.90	1.29	−44.40***	−8.43	−8.43
Home	(109.00)	(109.00)	(5.19)	(8.53)	(17.54)	(17.54)
	[37.48]	[37.48]	[37.83]	[37.39]	[38.06]	[38.06]
Obs.	1,458	1,458	1,698	1,924	2,121	2,121
Positive	75.64***	75.64***	49.02***	0.750	9.85	9.85
Leisure	(26.33)	(26.33)	(17.15)	(9.30)	(13.64)	(13.64)
	[32.13]	[32.13]	[31.55]	[32.14]	[31.76]	[31.76]
Obs.	1,458	1,458	1,698	1,924	2,121	2,121
Passive	−130.21	−130.21	−45.16	54.17**	−1.95	−1.95
Leisure	(109.00)	(109.00)	(39.07)	(23.88)	(17.02)	(17.02)
	[190.75]	[190.75]	[188.63]	[191.46]	[192.39]	[192.39]
Obs.	1,458	1,458	1,698	1,924	2,121	2,121
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. Panel A measures GPA among currently enrolled students and includes individual and time fixed effects. Its pre means and Obs. refer to the corresponding estimation sample; Obs. counts person-waves. Individual age is from 18 to 24. Panel B includes state and time fixed effects. The pre-treatment mean is reported in square brackets. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

In Panel B, Obs. reports respondent-day observations used in the corresponding time-use estimation.

Table A6: Robustness Results of Substance Ever Use

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Marijuana	0.072	0.071	0.075	0.075	0.075	0.075	0.075
	(0.011)	(0.013)	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
	[0.461]	[0.470]	[0.474]	[0.474]	[0.474]	[0.474]	[0.478]
Obs.	340	385	397	397	397	397	401
Cocaine	0.001	0.024	0.026	0.026	0.026	0.026	0.042
	(0.001)	(0.010)	(0.008)	(0.008)	(0.008)	(0.008)	(0.011)
	[0.090]	[0.098]	[0.094]	[0.094]	[0.094]	[0.094]	[0.093]
Obs.	340	385	397	397	397	397	401
Group A	0.014***	0.033***	0.032***	0.032***	0.032***	0.032***	0.032***
	(0.004)	(0.011)	(0.014)	(0.014)	(0.014)	(0.008)	(0.011)
	[0.161]	[0.165]	[0.168]	[0.168]	[0.168]	[0.168]	[0.166]
Obs.	340	385	397	397	397	397	401
Group B	−0.019	−0.010	−0.014*	−0.014*	−0.014*	−0.014*	−0.014*
	(0.004)	(0.010)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
	[0.148]	[0.140]	[0.144]	[0.144]	[0.144]	[0.144]	[0.146]
Obs.	340	385	397	397	397	397	401
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A6: Robustness Results of Substance Ever Use (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Marijuana	0.112***	0.112***	0.081***	0.041	0.002	0.002
	(0.032)	(0.032)	(0.032)	(0.027)	(0.054)	(0.054)
	[0.481]	[0.481]	[0.487]	[0.468]	[0.482]	[0.482]
Obs.	525	525	752	867	972	972
Cocaine	0.038***	0.038***	−0.007	0.044**	0.072**	0.072**
	(0.007)	(0.007)	(0.060)	(0.020)	(0.034)	(0.034)
	[0.088]	[0.088]	[0.086]	[0.085]	[0.089]	[0.089]
Obs.	525	525	752	867	972	972
Group A	0.598***	0.598***	−0.076	0.028	0.024**	0.024**
	(0.226)	(0.226)	(0.084)	(0.018)	(0.012)	(0.012)
	[0.163]	[0.163]	[0.182]	[0.179]	[0.173]	[0.173]
Obs.	525	525	752	867	972	972
Group B	0.578***	0.578***	0.045	0.085***	0.106***	0.106***
	(0.217)	(0.217)	(0.028)	(0.018)	(0.026)	(0.026)
	[0.145]	[0.145]	[0.128]	[0.121]	[0.122]	[0.122]
Obs.	525	525	752	867	972	972
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A7: Robustness Results of Substance Frequency in 30 Days

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Marijuana	0.095	2.14	2.16	2.16	2.16	2.16	2.16
	(0.005)	(0.786)	(0.744)	(0.744)	(0.744)	(0.744)	(0.744)
	[3.51]	[3.60]	[3.50]	[3.50]	[3.50]	[3.50]	[3.47]
Obs.	340	385	397	397	397	397	401
Cocaine	0.144	0.052	0.050	0.050	0.050	0.050	0.050
	(0.007)	(0.089)	(0.083)	(0.083)	(0.052)	(0.052)	(0.083)
	[0.052]	[0.062]	[0.060]	[0.060]	[0.060]	[0.060]	[0.060]
Obs.	340	385	397	397	397	397	401
Group A	0.856	0.217	0.320	0.320	0.320	0.320	0.320
	(0.050)	(0.866)	(0.785)	(0.785)	(0.785)	(0.657)	(0.657)
	[0.706]	[0.684]	[0.790]	[0.790]	[0.790]	[0.790]	[0.783]
Obs.	340	385	397	397	397	397	401
Group B	−0.319	−0.090	−0.099	−0.099	−0.099	−0.099	−0.099
	(0.005)	(0.193)	(0.195)	(0.195)	(0.195)	(0.187)	(0.187)
	[0.353]	[0.334]	[0.322]	[0.322]	[0.322]	[0.322]	[0.319]
Obs.	340	385	397	397	397	397	401
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A7: Robustness Results of Substance Frequency in 30 Days (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Marijuana	1.69***	1.69***	2.24***	1.26**	1.39**	1.39**
	(0.764)	(0.764)	(0.644)	(0.614)	(0.543)	(0.543)
	[3.55]	[3.55]	[4.40]	[4.37]	[4.55]	[4.55]
Obs.	525	525	751	866	971	971
Cocaine	−0.008	−0.008	0.060**	0.072***	0.116	0.116
	(0.031)	(0.031)	(0.026)	(0.025)	(0.072)	(0.072)
	[0.063]	[0.063]	[0.054]	[0.048]	[0.063]	[0.063]
Obs.	525	525	752	867	972	972
Group A	0.436	0.436	0.136	0.985**	0.889*	0.889*
	(0.526)	(0.526)	(0.401)	(0.488)	(0.490)	(0.490)
	[0.694]	[0.694]	[0.836]	[0.933]	[0.861]	[0.861]
Obs.	525	525	752	867	972	972
Group B	0.068	0.068	−0.029	−0.016	0.113	0.113
	(0.038)	(0.038)	(0.110)	(0.104)	(0.049)	(0.049)
	[0.287]	[0.287]	[0.207]	[0.186]	[0.170]	[0.170]
Obs.	525	525	752	867	972	972
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A8: Robustness Results of Substance Frequency in 12 Months

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Marijuana	-1.14^{***}	0.978	0.874	0.874	0.874	0.874	0.874
	(0.116)	(1.67)	(1.66)	(1.66)	(1.58)	(1.12)	(1.12)
	[5.51]	[6.04]	[5.93]	[5.93]	[5.93]	[5.93]	[5.88]
Obs.	340	385	397	397	397	397	401
Cocaine	0.445	0.456^{***}	0.440^{***}	0.440^{***}	0.440^{***}	0.440^{***}	0.440^{***}
	(0.042)	(0.050)	(0.076)	(0.076)	(0.076)	(0.073)	(0.073)
	[0.315]	[0.355]	[0.343]	[0.343]	[0.343]	[0.343]	[0.340]
Obs.	340	385	397	397	397	397	401
Group A	2.73^{***}	1.81^{***}	1.80^{***}	1.80^{***}	1.80^{***}	1.80^{***}	1.80^{***}
	(0.068)	(0.405)	(0.401)	(0.401)	(0.387)	(0.215)	(0.293)
	[1.97]	[1.89]	[2.06]	[2.06]	[2.06]	[2.06]	[2.05]
Obs.	340	385	397	397	397	397	401
Group B	-1.05^{***}	-0.621	-0.619	-0.619	-0.619	-0.619	-0.619
	(0.042)	(0.273)	(0.283)	(0.283)	(0.283)	(0.283)	(0.283)
	[0.868]	[0.820]	[0.791]	[0.791]	[0.791]	[0.791]	[0.784]
Obs.	340	385	397	397	397	397	401
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A8: Robustness Results of Substance Frequency in 12 Months (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Marijuana	2.75***	2.75***	3.67***	2.54***	2.82***	2.82***
	(0.923)	(0.923)	(0.472)	(0.667)	(1.02)	(1.02)
	[5.86]	[5.86]	[7.13]	[6.97]	[7.14]	[7.14]
Obs.	525	525	751	866	971	971
Cocaine	0.128	0.128	0.159	0.322***	0.589*	0.589*
	(0.162)	(0.162)	(0.109)	(0.086)	(0.284)	(0.284)
	[0.320]	[0.320]	[0.299]	[0.277]	[0.360]	[0.360]
Obs.	525	525	752	867	972	972
Group A	1.07	1.07	0.944	1.92***	1.82***	1.82***
	(0.571)	(0.571)	(0.974)	(0.518)	(0.359)	(0.359)
	[1.82]	[1.82]	[2.35]	[2.38]	[2.29]	[2.29]
Obs.	525	525	752	865	970	970
Group B	−0.163	−0.163	0.232	0.143	0.233	0.233
	(0.090)	(0.090)	(0.344)	(0.251)	(0.190)	(0.190)
	[0.709]	[0.709]	[0.648]	[0.585]	[0.535]	[0.535]
Obs.	525	525	752	867	972	972
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A9: Robustness Results of Substance Frequency in Lifetime

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Marijuana	0.190***	2.08	2.15	2.15	2.15	2.15	2.16
	(0.040)	(1.44)	(1.43)	(1.43)	(1.35)	(0.837)	(0.834)
	[9.23]	[9.71]	[9.88]	[9.88]	[9.88]	[9.88]	[9.92]
Obs.	340	385	397	397	397	397	401
Cocaine	−0.664	0.607	0.641	0.641	0.641	0.641	0.656
	(0.018)	(0.467)	(0.466)	(0.466)	(0.412)	(0.287)	(0.313)
	[1.10]	[1.19]	[1.14]	[1.14]	[1.14]	[1.14]	[1.13]
Obs.	340	385	397	397	397	397	401
Group A	0.084	1.63	1.73	1.73	1.73	1.73	1.73
	(0.073)	(1.18)	(1.14)	(1.14)	(1.00)	(0.732)	(0.811)
	[4.93]	[4.75]	[4.85]	[4.85]	[4.85]	[4.85]	[4.81]
Obs.	340	385	397	397	397	397	401
Group B	0.154	0.849	0.775	0.775	0.775	0.775	0.775
	(0.037)	(1.14)	(1.10)	(1.10)	(1.08)	(1.07)	(1.05)
	[1.56]	[1.48]	[1.46]	[1.46]	[1.46]	[1.46]	[1.45]
Obs.	340	385	397	397	397	397	401
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A9: Robustness Results of Substance Frequency in Lifetime (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Marijuana	4.15***	4.15***	2.98***	2.75***	3.06***	3.06***
	(0.868)	(0.868)	(0.692)	(0.857)	(0.648)	(0.648)
	[9.88]	[9.88]	[11.00]	[10.59]	[10.78]	[10.78]
Obs.	525	525	751	866	969	969
Cocaine	0.353	0.353	0.473	0.999***	1.74**	1.74**
	(0.456)	(0.456)	(0.335)	(0.194)	(0.672)	(0.672)
	[1.05]	[1.05]	[0.874]	[0.802]	[0.986]	[0.986]
Obs.	525	525	752	867	972	972
Group A	24.55***	24.55***	0.547	1.84**	1.80***	1.80***
	(9.06)	(9.06)	(1.47)	(0.841)	(0.427)	(0.427)
	[4.45]	[4.45]	[5.21]	[5.18]	[4.98]	[4.98]
Obs.	525	525	752	867	970	970
Group B	24.32***	24.32***	1.72	1.45***	2.22***	2.22***
	(8.79)	(8.79)	(1.16)	(0.484)	(0.641)	(0.641)
	[1.42]	[1.42]	[1.32]	[1.20]	[1.20]	[1.20]
Obs.	525	525	752	867	970	970
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Individual age is from 18 to 24. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A10: Robustness Results of Drink and Smoke

	(1) 2000	(2) 2001	(3) 2002	(4) 2003	(5) 2004	(6) 2005	(7) 2006
Panel A. Drink							
Ever	0.148***	0.143***	0.134***	0.134***	0.134***	0.139***	0.243***
	(0.003)	(0.006)	(0.017)	(0.017)	(0.016)	(0.016)	(0.047)
	[0.592]	[0.596]	[0.602]	[0.602]	[0.602]	[0.608]	[0.601]
Obs.	344	389	401	401	401	409	587
Quantity	−0.332	−0.299	−0.316	−0.316	−0.316	−0.274	0.689
	(0.322)	(0.373)	(0.339)	(0.339)	(0.339)	(0.211)	(0.466)
	[1.77]	[1.81]	[1.83]	[1.83]	[1.83]	[1.90]	[1.92]
Obs.	334	378	390	390	390	398	572
Panel B. Smoke							
Ever	−0.004	−0.014***	−0.011	−0.011	−0.011	−0.009	0.047***
	(0.001)	(0.003)	(0.011)	(0.011)	(0.011)	(0.010)	(0.015)
	[0.207]	[0.211]	[0.209]	[0.209]	[0.209]	[0.216]	[0.200]
Obs.	344	389	401	401	401	409	587
States	6	7	8	8	9	10	11

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator.

The quantity measure records drinks on drinking days, including verified non-drinking zeros; one drink or fewer is coded as one. Unknown or inconsistent responses remain missing. See Appendix A1.

Table A10: Robustness Results of Drink and Smoke (continued)

	(8) 2007	(9) 2008	(10) 2009	(11) **Baseline**	(12) 2011	(13) 2012
Panel A. Drink						
Ever	0.257^{***}	0.257^{***}	0.183^{***}	0.213^{***}	0.206^{***}	0.206^{***}
	(0.033)	(0.033)	(0.043)	(0.026)	(0.024)	(0.024)
	[0.595]	[0.595]	[0.603]	[0.632]	[0.620]	[0.620]
Obs.	604	604	892	1,019	1,166	1,166
Quantity	0.674	0.674	0.405	0.057	0.206	0.206
	(0.450)	(0.450)	(0.392)	(0.276)	(0.126)	(0.126)
	[1.95]	[1.95]	[1.95]	[2.05]	[2.04]	[2.04]
Obs.	588	588	871	994	1,139	1,139
Panel B. Smoke						
Ever	0.051^{***}	0.051^{***}	0.081^{***}	0.072^{***}	0.059^{***}	0.059^{***}
	(0.013)	(0.013)	(0.028)	(0.021)	(0.018)	(0.018)
	[0.200]	[0.200]	[0.226]	[0.215]	[0.219]	[0.219]
Obs.	604	604	890	1,012	1,158	1,158
States	12	12	13	15	17	17

Note: Each column is based on a different sample of states. State inclusion is determined by comparing the MML treatment year with the cutoff year, such that only states with an MML treatment year less than or equal to the cutoff year are included. The baseline year is set to 2010. The pre-treatment mean is reported in square brackets. Individual and time fixed effects are included. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator.

The quantity measure records drinks on drinking days, including verified non-drinking zeros; one drink or fewer is coded as one. Unknown or inconsistent responses remain missing. See Appendix A1.

Table A11: ATUS Activity Codes for Study and Leisure Time

Measure	Activity codes	Activities
Study at School	0601; 0602	Attending class; non-sport extracurricular school activities.
Study at Home	0603	Research or homework.
Leisure	1203	Relaxing and leisure.
Positive leisure	120301; 120309–120313; 120399; 1204	Relaxing and thinking; arts and personal-interest activities; other relaxing and leisure; out-of-home leisure, including museums and movies.
Passive leisure	120302–120308	Smoking, drug use, television, radio, and computer games, among other activities in these categories.

Notes: ATUS denotes the American Time Use Survey. Education-related time combines Study at School and Study at Home. The leisure measure follows Chu and Gershenson (2018); positive and passive leisure are the further classifications used in this paper. Four-digit codes denote broader activity categories; six-digit codes identify more detailed categories. Outcomes measure minutes per day.

Table A12: Placebo Tests for Reporting Bias Indicators

Outcome	True share (%)	Placebo mean (%)	Diff. (p.p.)	p-value
Marijuana	3.793	3.238	0.555	0.163
Cocaine	1.638	1.382	0.256	0.198
Amphetamines	1.724	1.460	0.264	0.281
Barbiturates	1.121	0.859	0.261	0.193
Diet pills	1.121	1.038	0.083	0.392
Steroids	0.948	0.967	-0.019	0.476
Tranquilizers	0.690	0.727	-0.037	0.525

Note: For each outcome, 1,000 placebo simulations are conducted by randomly reassigning RML legalization years across states. The table reports the observed share of individuals flagged by the reporting-bias indicator in the actual data, the mean share across placebo simulations, the difference between the observed share and the placebo mean, and the randomization p-value, defined as the proportion of placebo estimates that are at least as large as the observed value.

Table A13: Estimates of RML on BA Graduation Rate by Institution Type

	(1) Within 4 year	(2) Within 5 year	(3) Within 6 year
Panel A: Private vs Public			
Private	−0.031***	−0.020***	−0.026***
	(0.010)	(0.007)	(0.007)
	[0.510]	[0.601]	[0.620]
Obs.	1,626	1,591	1,591
Public	0.001	−0.001	−0.009
	(0.005)	(0.006)	(0.006)
	[0.288]	[0.471]	[0.532]
Obs.	1,125	1,115	1,115
Panel B: Carnegie Classification			
Doctoral	−0.003	−0.006	−0.012**
	(0.005)	(0.004)	(0.005)
	[0.451]	[0.626]	[0.675]
Obs.	665	655	655
Master	−0.009	0.013*	−0.023***
	(0.007)	(0.007)	(0.008)
	[0.351]	[0.479]	[0.521]
Obs.	1,070	1,035	1,035
Bacc/Assoc	−0.050***	−0.045***	−0.021
	(0.012)	(0.011)	(0.017)
	[0.510]	[0.592]	[0.612]
Obs.	619	619	619
Other	−0.003	0.007	−0.015
	(0.028)	(0.006)	(0.014)
	[0.414]	[0.526]	[0.554]
Obs.	387	387	387

Note: Panel A presents the estimated effects of RML on BA graduation rates within 4, 5, and 6 years, separately for private and public institutions. Panel B presents the estimated effects of RML on BA graduation rates within 4, 5, and 6 years by institutional Carnegie Classification. The pre-treatment mean is reported in square brackets. Institution and time fixed effects are included. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Each Obs. row reports institution–entry-cohort observations used in that subgroup estimation.

Table A14: Heterogeneity

	4 Year Institution			2 Year Institution
	(1) Within 4 year	(2) Within 5 year	(3) Within 6 year	(4) Within 2 year
Panel A: Gender				
Female	−0.014**	−0.003	−0.016	−0.023***
	(0.005)	(0.007)	(0.011)	(0.004)
	[0.455]	[0.578]	[0.611]	[0.347]
Obs.	2,741	2,696	2,696	1,832
Male	−0.019**	−0.019***	−0.016***	−0.015*
	(0.008)	(0.006)	(0.004)	(0.008)
	[0.375]	[0.511]	[0.553]	[0.321]
Obs.	2,728	2,681	2,681	1,806
Panel B: Race				
White	−0.025***	−0.018***	−0.020***	−0.004
	(0.007)	(0.005)	(0.004)	(0.012)
	[0.447]	[0.571]	[0.602]	[0.363]
Obs.	2,742	2,697	2,697	1,796
Black	−0.005	0.002	−0.010	−0.002
	(0.017)	(0.015)	(0.022)	(0.008)
	[0.298]	[0.429]	[0.467]	[0.281]
Obs.	2,595	2,555	2,562	1,315
Hispanic	−0.007	−0.012***	−0.036*	−0.012
	(0.014)	(0.009)	(0.019)	(0.008)
	[0.368]	[0.502]	[0.540]	[0.332]
Obs.	2,683	2,634	2,648	1,706
Asian	−0.021	−0.037***	−0.039**	0.008
	(0.015)	(0.013)	(0.016)	(0.023)
	[0.428]	[0.568]	[0.609]	[0.382]
Obs.	2,600	2,537	2,509	1,597

Note: Panel A presents the estimated effects of RML on graduation rates, separately for gender. Panel B presents the estimated effects of RML on graduation rates within 4, 5, and 6 years by race. The pre-treatment mean is reported in square brackets. Institution and time fixed effects are included. Not-yet-treated groups are used as the control group. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.
Each Obs. row reports institution–entry-cohort observations used in that subgroup estimation.

Table A15: Estimates of RML Effects on Other Drugs Use

	Ever	Frequency		
		In 30 days	In 12 months	In Lifetime
	(1)	(2)	(3)	(4)
Cocaine	0.044**	0.072***	0.322***	0.999***
	(0.020)	(0.025)	(0.086)	(0.194)
	[0.085]	[0.048]	[0.277]	[0.802]
Obs.	867	867	867	867
Group A	0.028	0.985**	1.92***	1.84**
	(0.018)	(0.488)	(0.518)	(0.841)
	[0.179]	[0.933]	[2.38]	[5.18]
Obs.	867	867	865	867
Group B	0.085***	−0.016	0.143	1.45***
	(0.018)	(0.104)	(0.251)	(0.484)
	[0.121]	[0.186]	[0.585]	[1.20]
Obs.	867	867	867	867

Note: Individual and time fixed effects are included. The not-yet treated groups are used as the control group. The pre-treatment mean is reported in square brackets. Individual age is from 18 to 24. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator. Drug-use frequency measures are midpoint-coded scores; unknown responses remain missing, and composite scores require all components to be observed.

Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Some items include prescribed use.

Table A16: Estimates of RML Effects on Drink and Smoke

	Ever Drink	Drinks per drinking day	Ever Smoke
	(1)	(2)	(3)
ATT	0.213***	0.057	0.072***
	(0.026)	(0.276)	(0.021)
Pre Means	0.632	2.05	0.215
Obs.	1,019	994	1,012

Note: Individual and time fixed effects are included. The not-yet treated groups are used as the control group. Individual age is from 18 to 24. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Pre means use untreated observations in the corresponding estimation sample, including never-treated units. Obs. counts person-wave observations used by the estimator.

The quantity measure records drinks on drinking days, including verified non-drinking zeros; one drink or fewer is coded as one. Unknown or inconsistent responses remain missing. See Appendix A1.

Table A17: Robustness Results of Inconsistent Substance and Health Behavior

	Ever		Frequency			
			In 30 days	In 12 months	In Lifetime	Drinks per drinking day
	(1)	(2)	(3)	(4)	(5)	(6)
Marijuana	0.060***	0.050***	1.17*	2.67***	3.21***	
	(0.015)	(0.015)	(0.603)	(0.670)	(0.833)	
	[0.486]	[0.450]	[4.48]	[7.19]	[10.64]	
Obs.	867	803	802	802	802	
Cocaine	0.040*	0.050***	0.087***	0.338***	0.955***	
	(0.021)	(0.013)	(0.026)	(0.098)	(0.191)	
	[0.086]	[0.079]	[0.046]	[0.278]	[0.797]	
Obs.	867	858	858	858	858	
Group A	−0.020	0.031*	0.968**	1.88***	0.248	
	(0.026)	(0.017)	(0.480)	(0.229)	(0.650)	
	[0.212]	[0.134]	[0.854]	[2.06]	[4.32]	
Obs.	867	777	777	775	777	
Group B	0.026	0.033	0.225	0.036	0.760***	
	(0.018)	(0.014)	(0.159)	(0.160)	(0.156)	
	[0.157]	[0.070]	[0.036]	[0.206]	[0.454]	
Obs.	867	780	780	780	780	
Drink	0.137***	0.134***				−0.084
	(0.022)	(0.016)				(0.110)
	[0.676]	[0.652]				[2.10]
Obs.	1,019	883				865
Smoke	0.052**	0.040***				
	(0.027)	(0.016)				
	[0.235]	[0.195]				
Obs.	1,012	952				

Note: Column (1) codes all ever-use reports after the first positive report as use. Column (2) excludes respondents with reversals in ever-use reports; columns (3)–(6) report frequency and quantity estimates after the corresponding exclusions. Respondents are aged 18–24. Individual and time fixed effects are included; not-yet-treated units are controls. Parentheses contain standard errors; square brackets contain untreated means from the corresponding estimation sample, including never-treated units. Obs. counts estimation person-waves. Standard errors and stars use state-clustered wild bootstrap inference; *, **, and *** denote significance at 10%, 5%, and 1%.

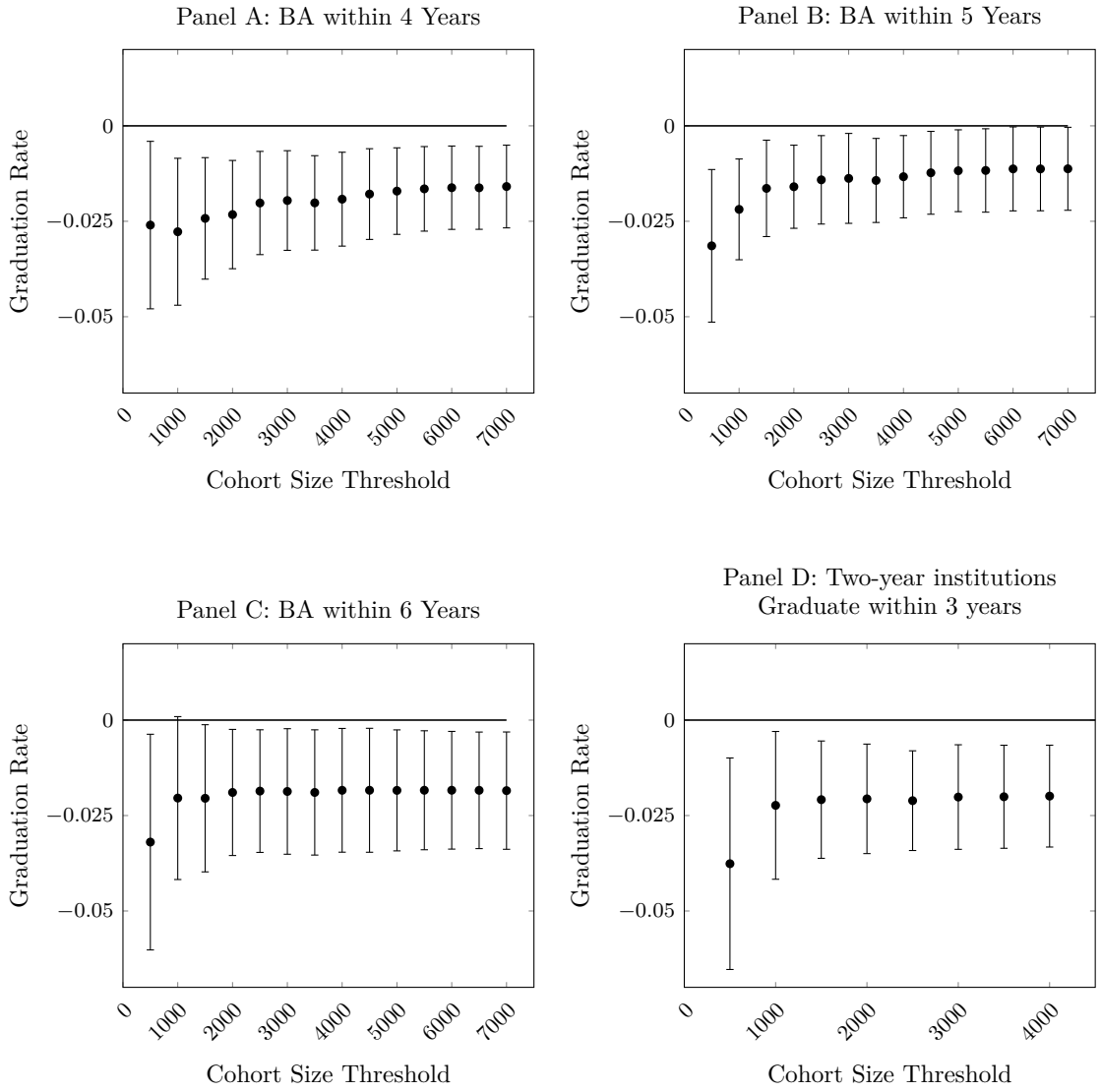
Group A comprises amphetamines, barbiturates, and tranquilizers; some items include prescribed use. Drug-frequency scores require valid responses for every component. Group B comprises diet pills and steroids. Drink quantity records drinks on drinking days with verified non-drinking zeros; one drink or fewer is coded as one. Unknown or inconsistent responses are missing. See Appendix A1.

Table A18: Estimates of MML on Education Attainment

	4 Year Institution		
	Within 4 year	Within 5 year	Within 6 year
	(1)	(2)	(3)
ATT	-0.031^{***}	0.028^{**}	0.026
	(0.012)	(0.013)	(0.026)
Pre mean	0.295	0.471	0.517
Obs.	440	421	435

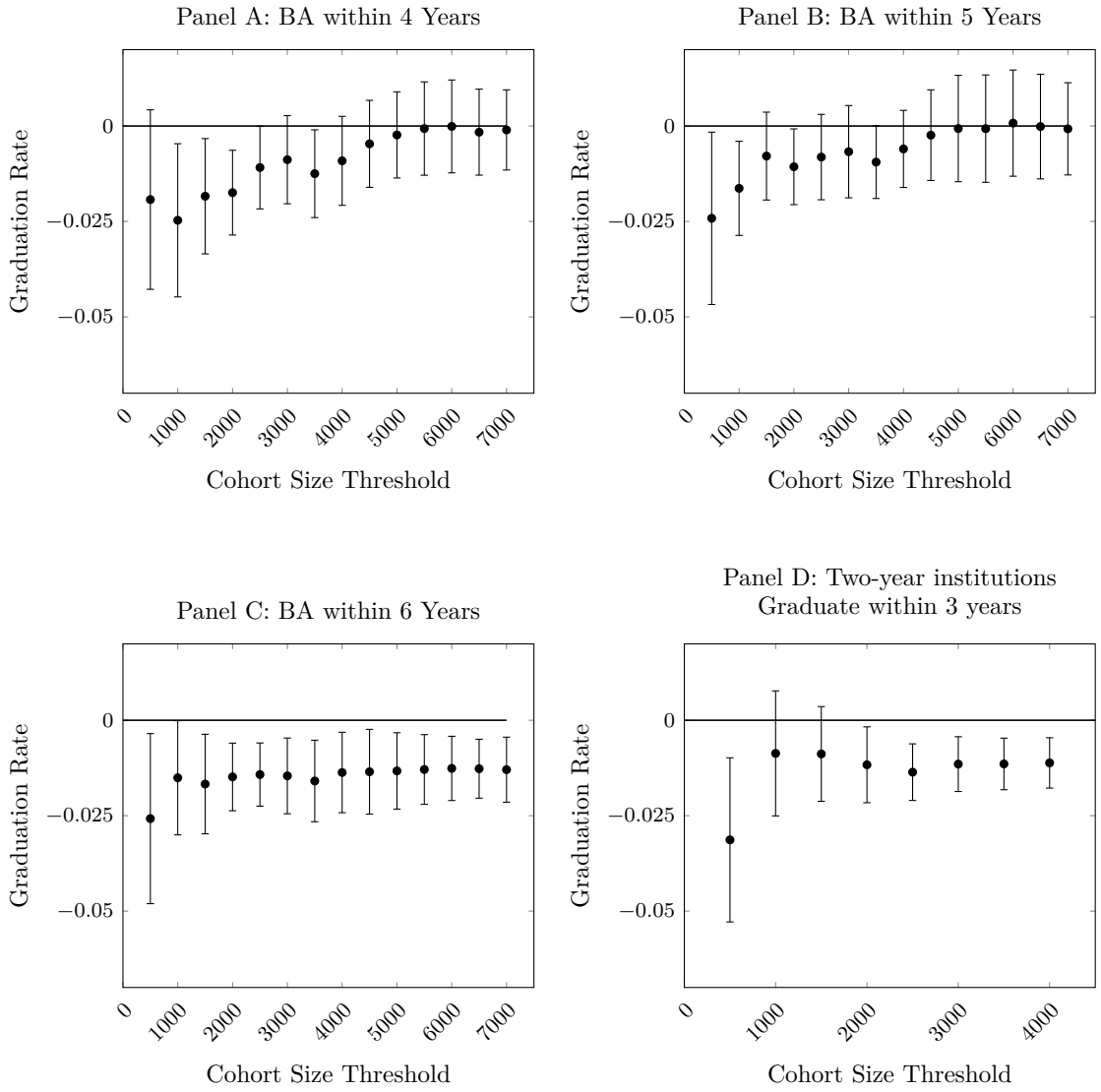
Note: The table reports the estimated effects of MML on BA completion within four, five, and six years at four-year institutions, using IPEDS data from 1997 to 2012 and the sample-screening procedures used in the RML analysis. Estimates use the Callaway and Sant'Anna (2021) estimator with not-yet-treated comparison groups. Standard errors (in parentheses) and significance stars are based on wild bootstrap inference with clustering at the state level. Pre mean is the pre-treatment mean of the outcome. Obs. reports institution–entry-cohort observations used in each estimation. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Figure A1: DiD Estimates for Graduation Rates by Cohort Size



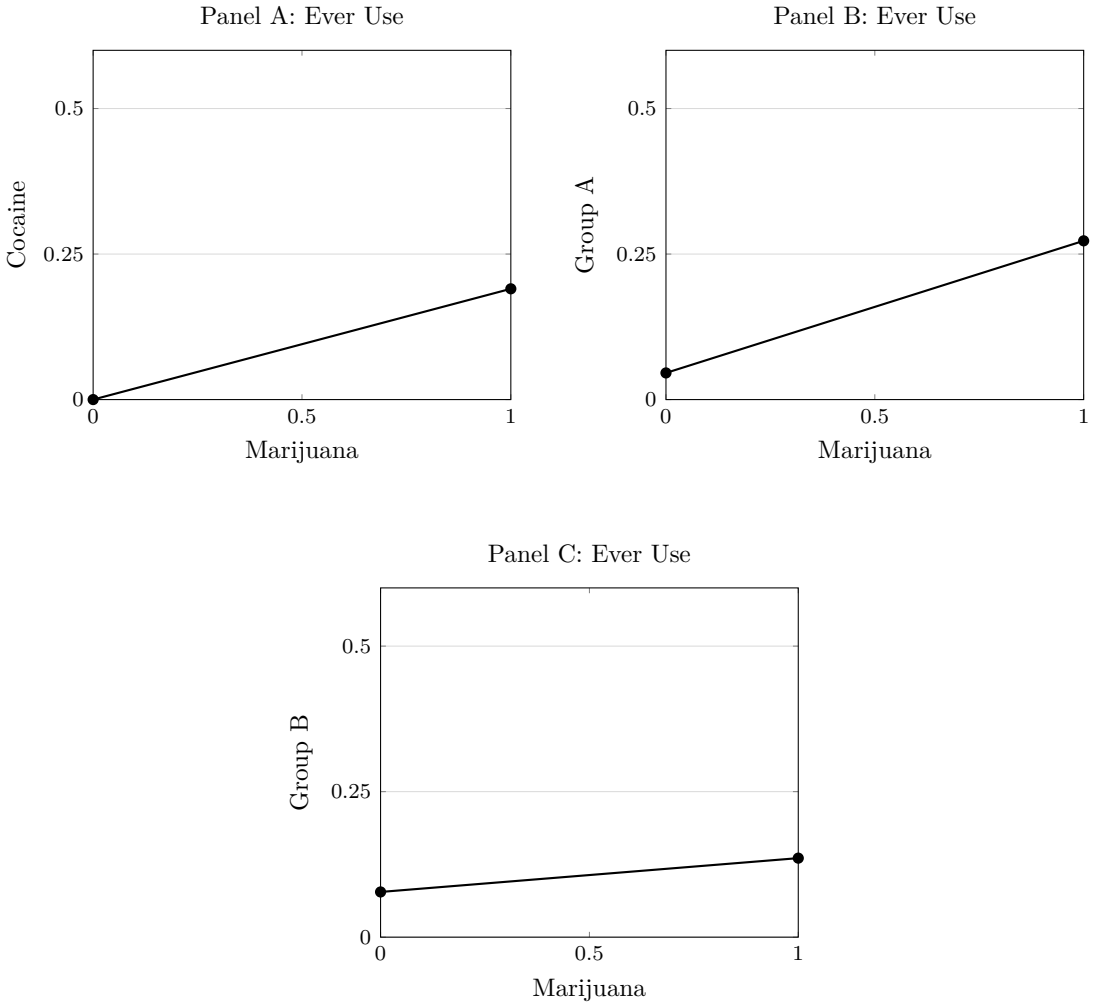
Note: Points report the Callaway and Sant'Anna (2021) estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level. The specifications follow Equation 1, with samples restricted by cohort size. An x-axis value of "i" where i = 500,1000,1500,...7000 indicates an estimate from a DiD analysis comparing graduation outcomes in treated and comparison cohort with less than i individuals.

Figure A2: DiD Estimates for Graduation Rates by Cohort Size, Weighted by Cohort Size



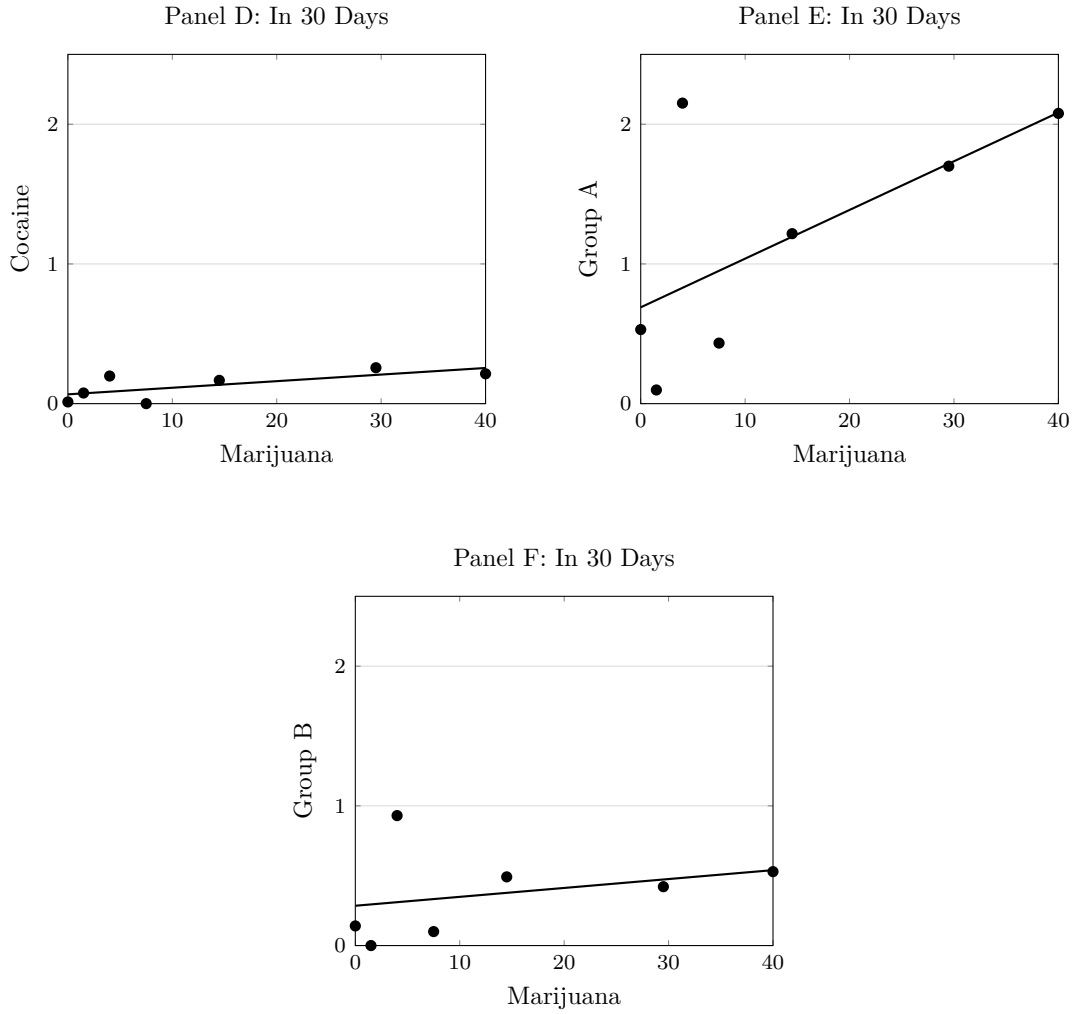
Note: Points report the Callaway and Sant'Anna (2021) estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level. The specifications follow Equation 1, with samples restricted by cohort size and weighted by cohort size. An x-axis value of "i" where i = 500,1000,1500,...7000 indicates an estimate from a DiD analysis comparing graduation outcomes in treated and comparison cohort with less than i individuals.

Figure A3: Relationship between Marijuana and Other Substance Use



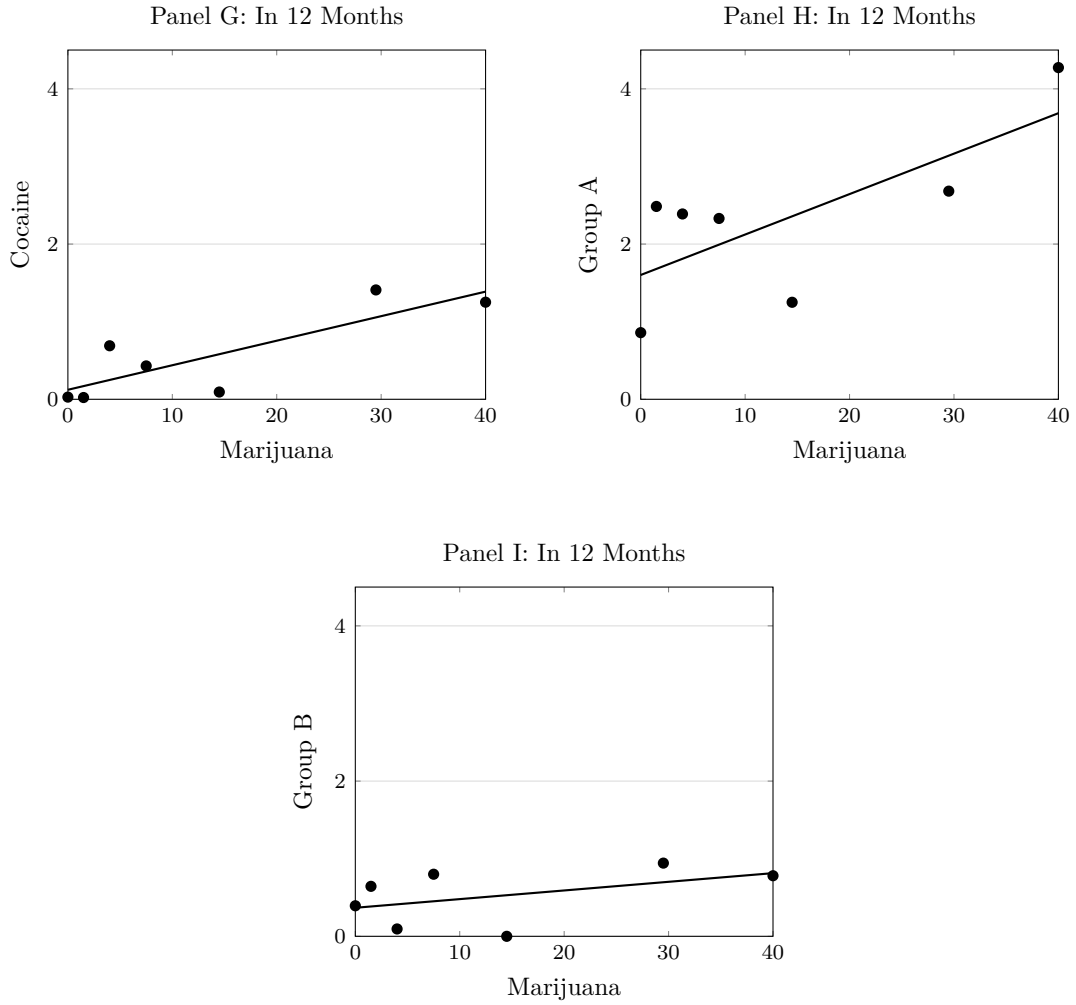
Note: Each panel plots the mean other drug-use outcome against the corresponding marijuana-use measure. Dots represent observed group means; lines fit these means with equal weight per nonempty marijuana category. Frequency measures are midpoint scores, summed across substances for the composite outcomes. Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Reported use may include prescribed use. The data are from PSID.

Figure A3: Relationship between Marijuana and Other Substance Use (continued)



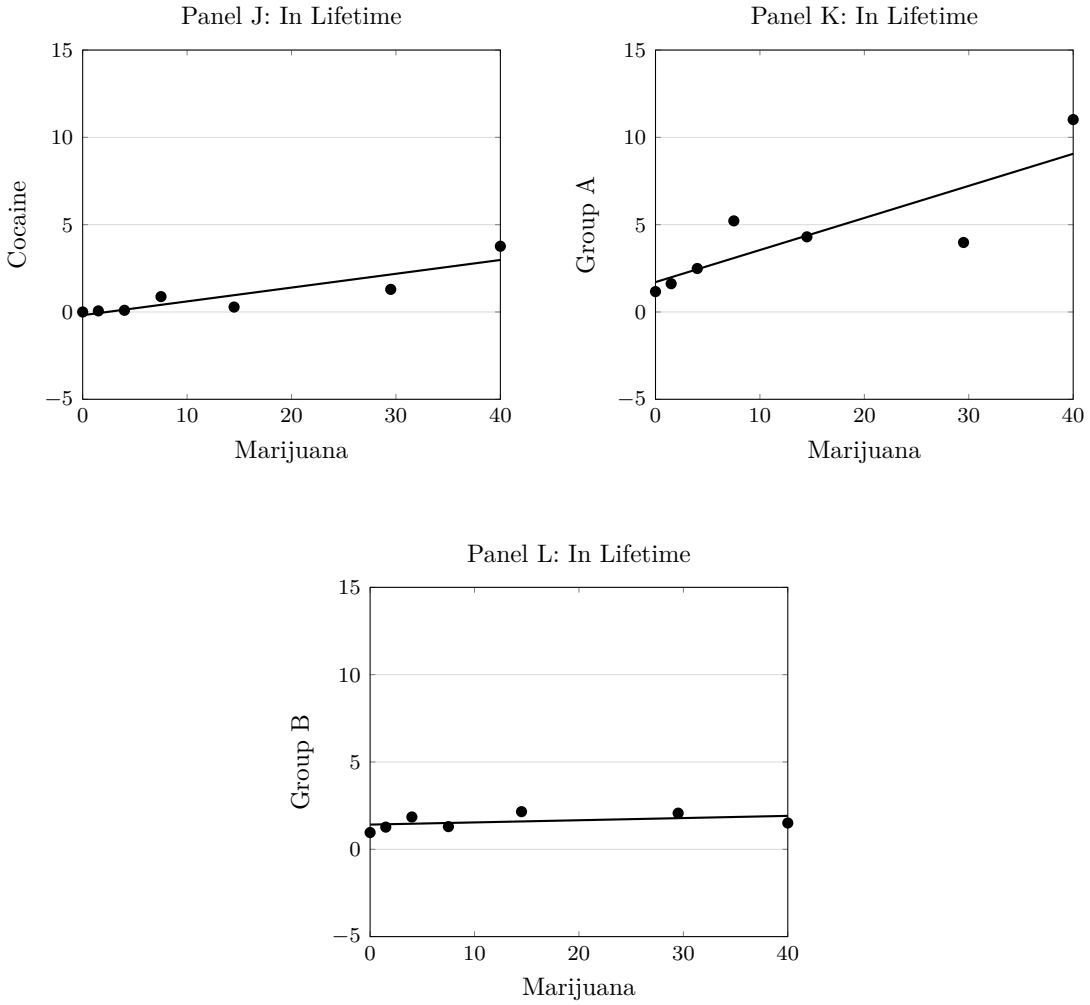
Note: Each panel plots the mean other drug-use outcome against the corresponding marijuana-use measure. Dots represent observed group means; lines fit these means with equal weight per nonempty marijuana category. Frequency measures are midpoint scores, summed across substances for the composite outcomes. Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Reported use may include prescribed use. The data are from PSID.

Figure A3: Relationship between Marijuana and Other Substance Use (continued)



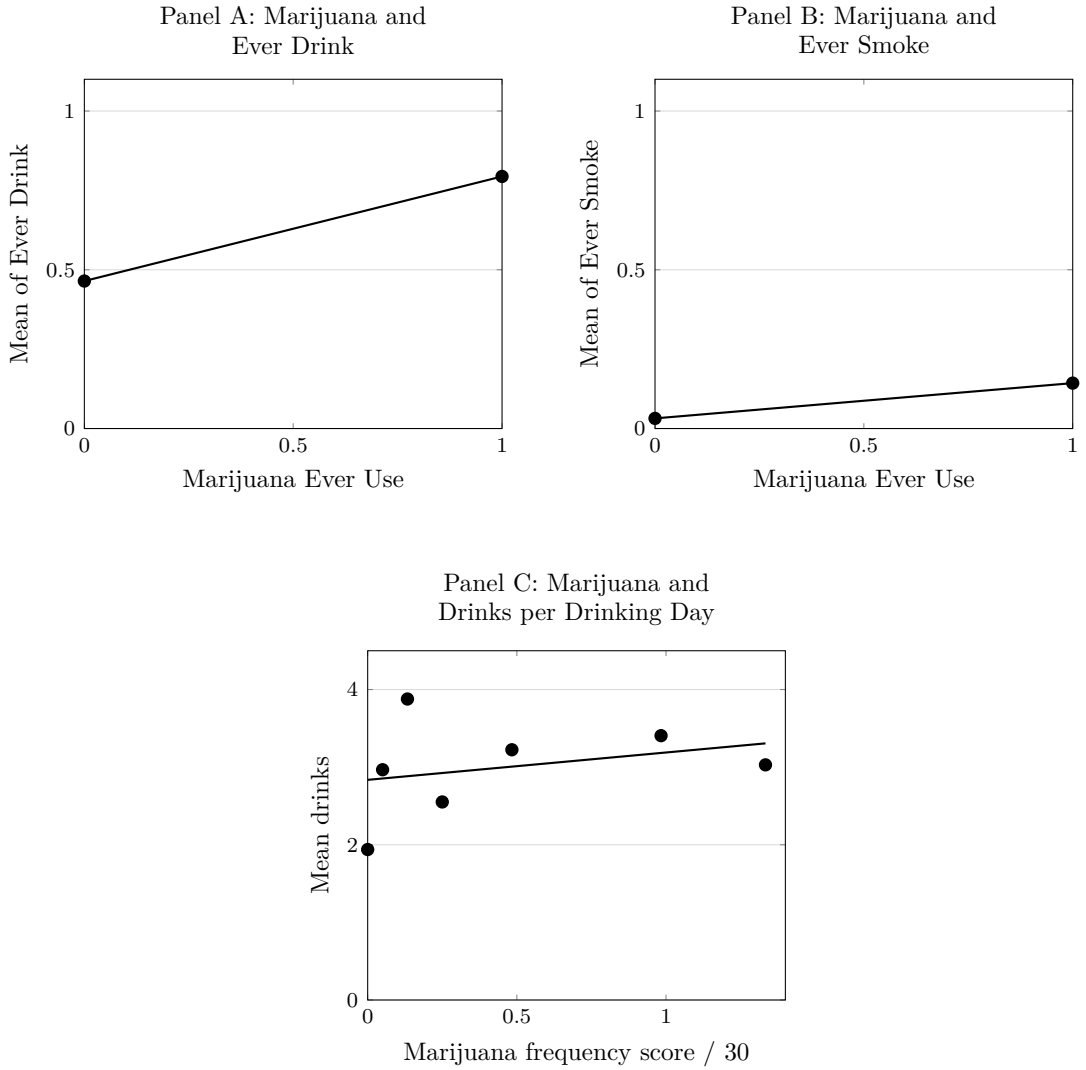
Note: Each panel plots the mean other drug-use outcome against the corresponding marijuana-use measure. Dots represent observed group means; lines fit these means with equal weight per nonempty marijuana category. Frequency measures are midpoint scores, summed across substances for the composite outcomes. Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Reported use may include prescribed use. The data are from PSID.

Figure A3: Relationship between Marijuana and Other Substance Use (continued)



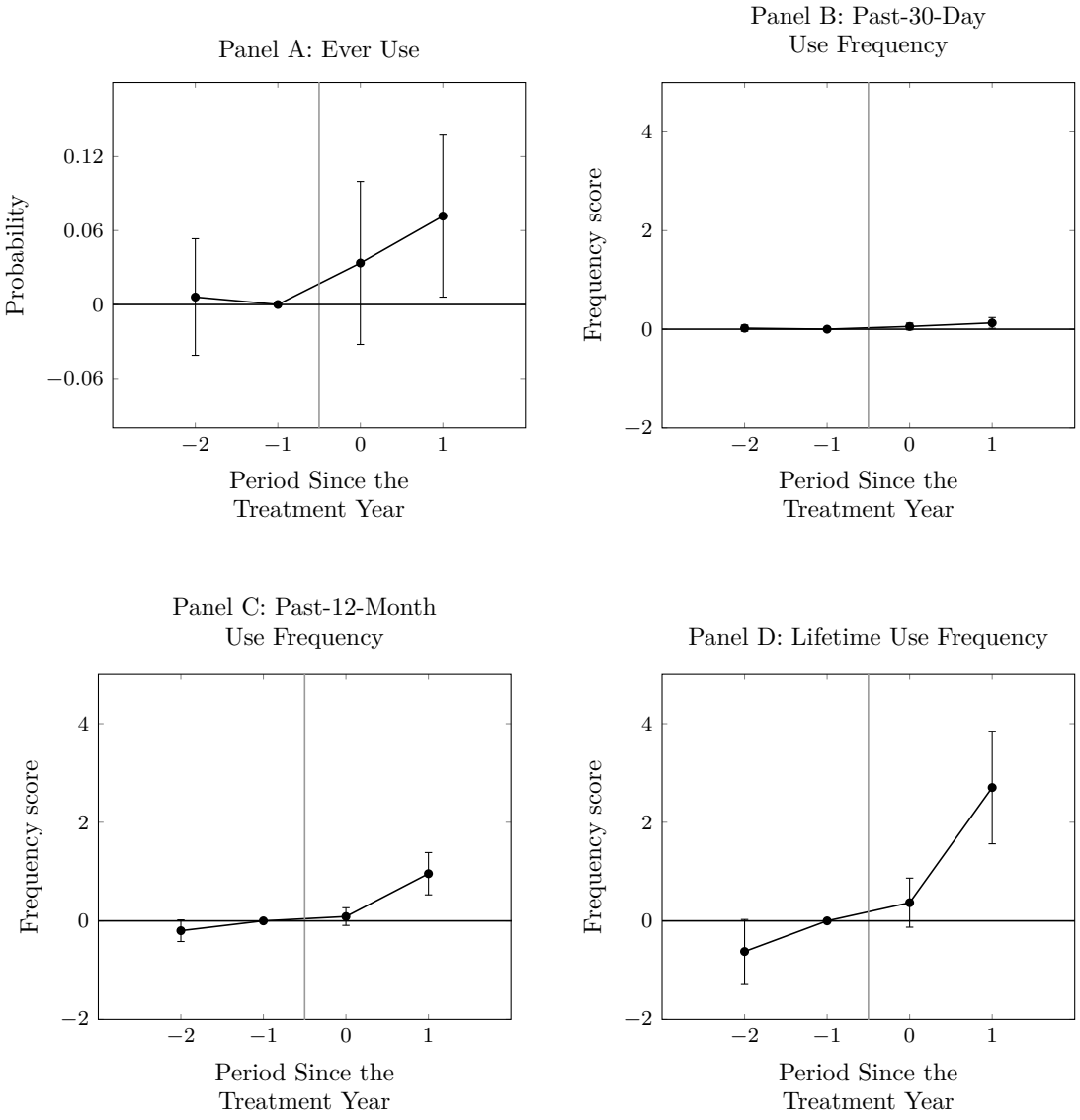
Note: Each panel plots the mean other drug-use outcome against the corresponding marijuana-use measure. Dots represent observed group means; lines fit these means with equal weight per nonempty marijuana category. Frequency measures are midpoint scores, summed across substances for the composite outcomes. Group A comprises amphetamines, barbiturates, and tranquilizers. Group B comprises diet pills and steroids. Reported use may include prescribed use. The data are from PSID.

Figure A4: Relationship between Marijuana and Drink/Smoke



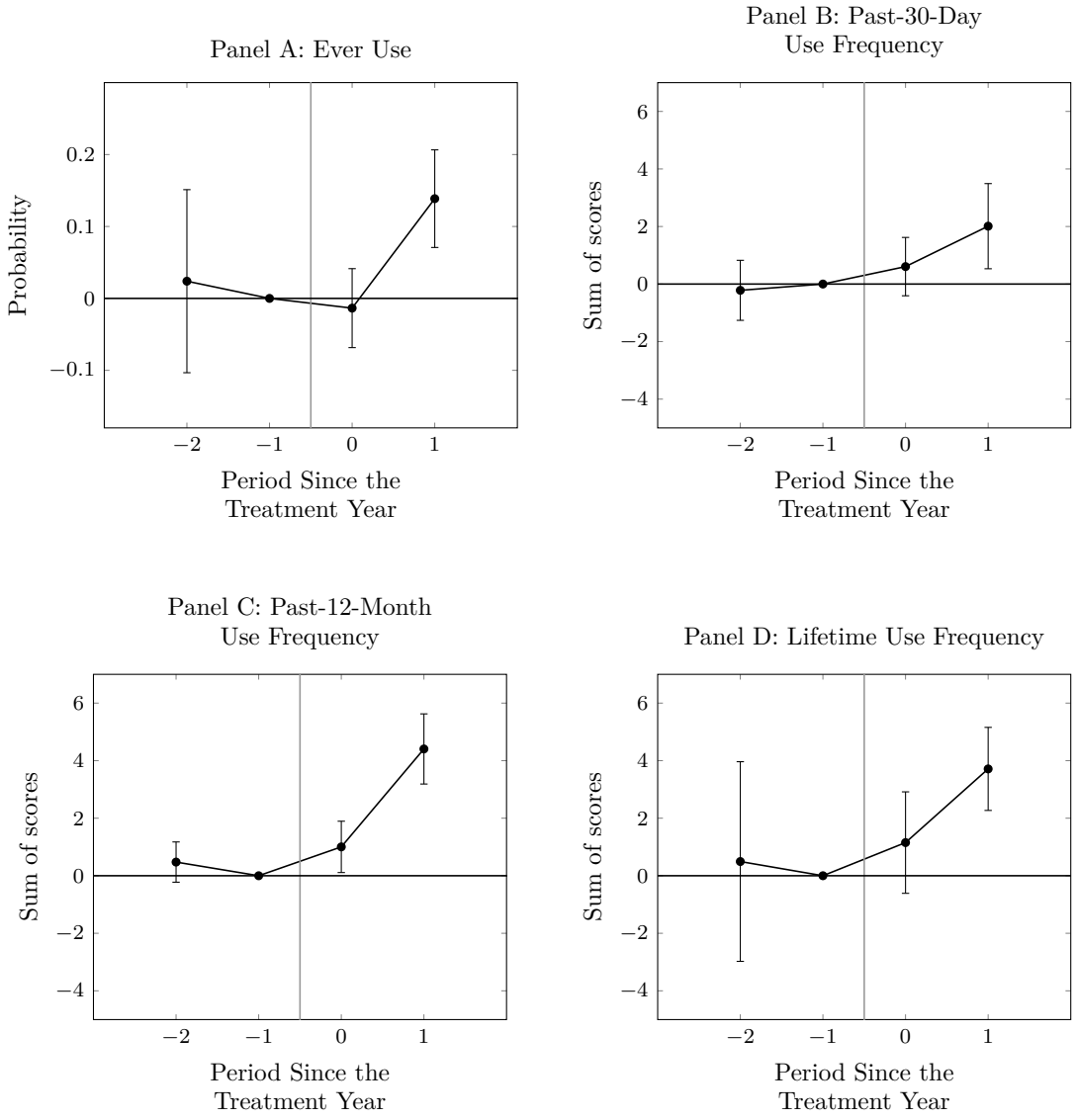
Note: Each panel plots the mean drink/smoke outcome against the corresponding marijuana-use measure. Dots represent observed group means, and the line indicates the fitted linear trend. Panel C uses reported drinks per drinking day, including zeros for respondents known not to have drunk. Its horizontal axis divides the past-30-day marijuana frequency score by 30 for display; it does not measure calendar-day consumption. The data is from PSID.

Figure A5: RML Event Study Estimates for Cocaine Use



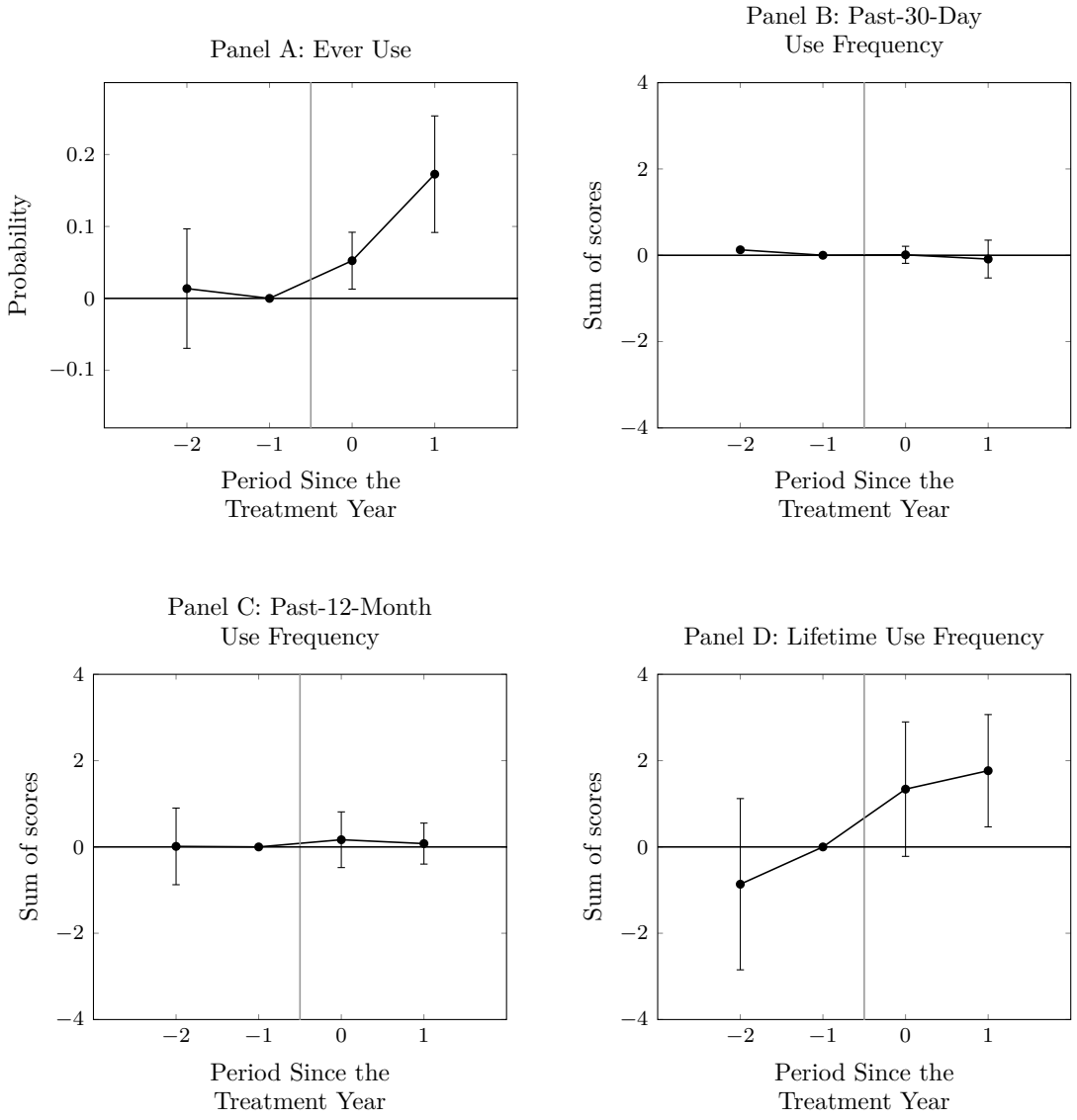
Note: Panel A present the event-study estimates for cocaine ever use, while panels B–D present the corresponding event-study estimates for cocaine use frequency. All specifications include individual and period fixed effects. Each period represents two years. Not-yet-treated groups are used as the control group. The sample is restricted to individuals ages 18–24. Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level.

Figure A6: RML Event Study Estimates for Group A Substance Use



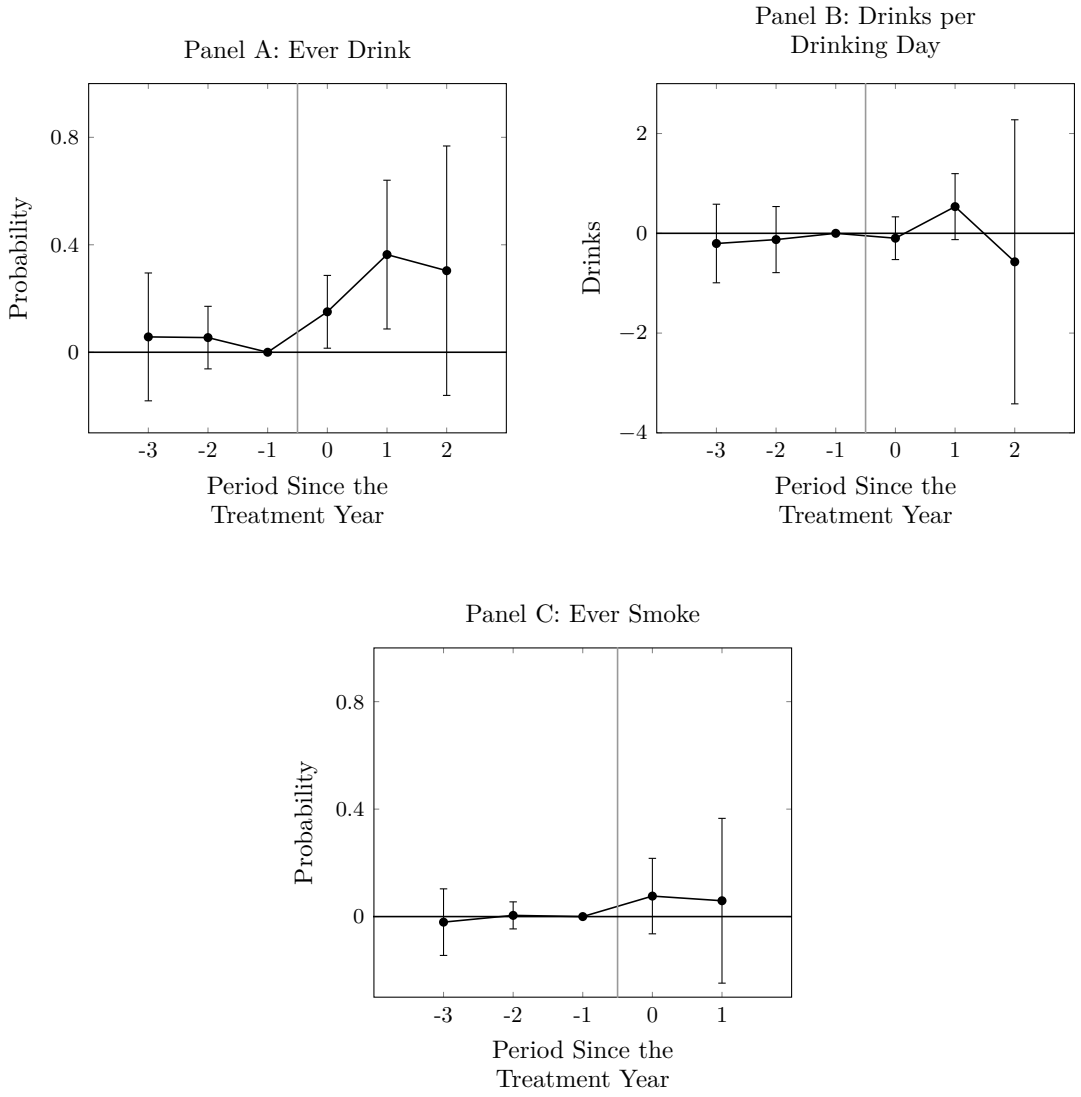
Note: Group A comprises amphetamines, barbiturates, and tranquilizers; reported use may include prescribed use. Panel A present the event-study estimates for Group A ever use, while panels B–D present the corresponding event-study estimates for use frequency. All specifications include individual and period fixed effects. Each period represents two years. Not-yet-treated groups are used as the control group. The sample is restricted to individuals ages 18–24. Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level.

Figure A7: RML Event Study Estimates for Group B Use



Note: Group B comprises diet pills and steroids. Panel A presents the event-study estimates for ever use, while panels B–D present the corresponding event-study estimates for use frequency. All specifications include individual and period fixed effects. Each period represents two years. Not-yet-treated groups are used as the control group. The sample is restricted to individuals ages 18–24. Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level.

Figure A8: RML Event Study Plots for Drink and Smoke



Note: Panel B uses reported drinks per drinking day, including zeros for respondents known not to have drunk. The not-yet treated groups are used as the control group. Individual age is from 18 to 24. Points report the Callaway and Sant'Anna (2021) event-study estimates, and bars show 95% confidence intervals based on standard errors clustered at the state level.